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Improved Wall Function Method for Hypersonic Reacting Turbulent Boundary Layers

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The chemical reaction and cold wall effects in hypersonic reacting turbulent boundary layers (HRTBLs) cause the failure of the existing wall function method. Therefore, this paper proposes an improved wall function method to simulate skin friction c_f and wall heat flux q_w in HRTBLs accurately and efficiently. Firstly, the temperature–velocity relation is extended to a generalized enthalpy–velocity relation to comprehensively involve the chemical reaction and cold wall effects. Then, a new unified velocity law of the wall for the inner layer of HRTBLs is theoretically derived. Finally, an improved wall function method is designed with the new laws of the wall. Numerical experiments of typical HRTBLs show that, compared with the existing wall function method, the improved wall function method can significantly reduce the errors of c_f and q_w from 8.0% and 10.2% to 0.4% and 0.5%, respectively, when the near-wall grids are relaxed to the viscous sublayer, while from 11.0% and 15.3% to 1.2% and 5.8%, respectively, when the near-wall grids are further relaxed to the logarithmic layer, bringing 2.7 and 15 times enhancement of the simulation efficiency. Moreover, by introducing the reference point, the improved wall function method removes the boundary layer edge quantities while ensuring accurate results.

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Nomenclature

B	=	6.0; logarithmic layer intercept
c_f	=	$\tau_w/(0.5\rho_\infty u_\infty^2)$; skin friction coefficient
c_p	=	specific heat at constant pressure, J/(kg · K)
D	=	mass diffusivity coefficient, m^2/s
E	=	total energy per unit mass, J/kg
H	=	total enthalpy per unit mass, J/kg
h	=	static enthalpy of the gas mixture, J/kg
h_s	=	static enthalpy of species s, J/kg
M_s	=	molar mass of species s
Ma	=	Mach number
Pr	=	Prandtl number
Pr_t	=	0.9; turbulent Prandtl number
p	=	pressure, Pa
q_w	=	$\lambda(\partial T/\partial n)\|_w$; wall heat flux, W/m^2
r	=	$Pr^{1/3}$; recovery factor
r_g	=	generalized recovery factor
Sc_t	=	0.9; turbulent Schimdt number
St	=	$q_w/(\rho_e u_e c_p(T_r - T_w))$; Stanton number
T	=	temperature, K
T_r	=	$T_e + ru_e^2/(2c_p)$; adiabatic wall temperature, K
u_i	=	velocity in the i direction, m/s
u_τ	=	$\sqrt{\tau_w/\rho_w}$; friction velocity, m/s
Y_s	=	mass fraction of species s
β	=	parameter model's heat transfer effect
γ	=	ratio of specific heat
Γ	=	parameter model's compressibility effect
Θ	=	T_w/T_r; wall temperature ratio
κ	=	0.41; von Kármán constant
λ	=	molecular thermal conductivity, W/(m · K)
μ	=	molecular viscosity, kg/(m · s)
μ_t	=	eddy viscosity, kg/(m · s)
ρ	=	density, kg/m^3
τ	=	shear stress, N/m^2
$\dot{\omega}$	=	mass production per unit volume per unit time, $kg/(m^3 \cdot s)$

Subscripts

e	=	boundary layer edge quantities
e^-	=	reference point quantities
s	=	species
w	=	wall quantities
1	=	quantities at the first grid point off the wall
∞	=	freestream quantities

Superscript

$+$	=	quantities in inner wall units

I. Introduction

THE fast and accurate prediction of the aerodynamic/aerothermal environment of hypersonic vehicles is a critical issue in computational fluid dynamics (CFD), and it will significantly accelerate the design process of hypersonic vehicles. However, the velocity and temperature in hypersonic turbulent boundary layers (HTBLs) vary sharply near the wall [1,2], which will be further aggravated with the increase of Mach number. Moreover, the massive amount of kinetic energy in hypersonic flows will be transformed into internal energy due to the effect of the shock wave and molecular viscosity, and the dissociation/recombination chemical reactions may occur in the high-temperature air of HTBLs. These physical phenomena will all bring great difficulties in fast and accurate predictions of skin friction c_f and wall heat flux q_w. Available studies [3–5] have shown that the numerical results of c_f and q_w are highly sensitive to the distribution of grid points near the wall, especially the distance of the first grid point off the wall Δy_1. To accurately simulate c_f and q_w of hypersonic vehicles, Δy_1 needs to be extremely small, generally 10^{-6} m or even smaller to satisfy the relation $\Delta y_1^+ = \rho_w u_\tau \Delta y_1/\mu_w < 1$, leading to a huge computational consumption. For three-dimensional complicated configurations, the dense mesh will limit the numerical integration timestep, and the overall computational consumption will be even unacceptable when the complex chemical reactions in HTBLs are simulated. An effective approach to relax the near-wall

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grid dependence of Reynolds-averaged Navier–Stokes equations (RANS) simulation of compressible turbulent boundary layers in engineering applications is to use the wall function method [6], which directly uses the velocity and temperature laws of the wall to represent their distributions, instead of resolving the near-wall region of turbulent boundary layers, and then obtain c_f and q_w. Because the velocity and temperature in the near-wall region are obtained from the laws of the wall, the first gird point off the wall can be placed further away from the wall, which reduces the mesh density and enlarges the numerical integration timestep. Therefore, the wall function method can effectively improve the computational efficiency of RANS simulation. Besides, the wall function method can also be used to improve the computational efficiency for the large-eddy simulation (LES) of compressible turbulent boundary layer flows [7,8]. Recently, Fu and Moin [8] proposed a near-wall model based on an inverse velocity transformation for LES, which is a typical representative.

This paper focuses on the wall function method for RANS simulations. Nichols and Nelson [9] first proposed a compressible wall function method by introducing the White compressible velocity law of the wall [10] and Walz temperature–velocity relation [11,12]. This wall function method involves compressibility and heat transfer effects in high-speed flows and is unified valid in the inner layer of turbulent boundary layers, making it possible to use the wall function method to obtain c_f and q_w in RANS simulation of supersonic turbulent boundary layers. Subsequently, Gao et al. [13] revised the parameters in the laws of the wall used by Nichols and Nelson according to theoretical analysis and numerical experiments and further achieved a more accurate coupling of the wall function method with CFD code. According to Gao et al.'s reports [13], this revised wall function method predicts c_f and q_w accurately for supersonic turbulent boundary layers on the flat plate when $\Delta y_1^+ < 400$. As for the application of separation flows, it was found from the numerical cases of supersonic cavity flow and hypersonic axisymmetric compression corner flow that the compressible velocity law of the wall originally established based on fully developed attached turbulent boundary layers is approximately valid for the near-wall region in separation flow, ensuring that the wall function method can provide reliable c_f and q_w inside the separation flows.

However, the existing compressible wall function methods [9,13–16] can only be applied to supersonic turbulent boundary layers with freestream Mach number less than five due to the fact that the laws of the wall they used will appear with significant errors in HTBLs. In fact, there will be more complex physical phenomena in HTBLs, mainly manifested in two aspects. On the one hand, the total temperature of hypersonic flows is extremely high, whereas the real wall temperature of hypersonic vehicles is usually significantly lower than the adiabatic wall temperature. At this time, the wall temperature ratio Θ may be less than 0.1, and there will be a colder wall on HTBLs than supersonic flows. Many studies have shown that the traditional velocity law of the wall and Walz relation show significant deviation from DNS results under these severe cold wall conditions [17–19]. On the other hand, the massive amount of kinetic energy in hypersonic flows will be transformed into internal energy due to the shock wave and molecular viscosity when hypersonic vehicles fly at an extremely high Mach number (e.g., $Ma_\infty > 15$). Hence, the dissociation/recombination chemical reactions may occur in the high-temperature air, which will significantly affect the temperature distribution in hypersonic reacting turbulent boundary layers (HRTBLs), and the traditional temperature–velocity relation, which does not involve the chemical reaction effect, will be no longer valid [20,21]. In summary, the complex effects of the chemical reactions and severe cold wall in HRTBLs will cause the failure of the existing laws of the wall as well as the existing wall function method, so it is necessary to derive the new laws of the wall and further establish an improved wall function method for HRTBLs.

At present, there have been several studies on the temperature–velocity relation under the cold wall condition [2,17–19], among which the study of Zhang et al. [17] is the most representative. Zhang et al. introduced the concept of generalized recovery factor r_g into the traditional Walz relation to consider the cold wall effect, and the newly proposed temperature–velocity relation collapses obviously better with the DNS results of HTBLs under cold wall conditions. However, this relation is derived based on the calorically prefect-gas assumption, which cannot be directly applied to HRTBLs. Actually, Gao et al. [22] first proposed to extend the Walz temperature–velocity relation to a generalized relation of enthalpy and velocity to involve the chemical reaction effect in their study on the wall function method for RANS simulation of supersonic combustion turbulent boundary layers. This approach could be adopted to extend Zhang's temperature–velocity relation to a generalized enthalpy–velocity relation to construct a new wall function, which is the work to be carried out in this paper. In terms of the velocity law of the wall, the existing compressible wall function is mainly established for the viscous sublayer and logarithmic layer [9,13–16]. In the viscous sublayer, the incompressible velocity law of the wall, that is, $u^+ = y^+$, is directly followed. However, Carvin et al. [23] and Trettel and Larsson [24] reported that $u^+ = y^+$ is already invalid in HTBLs with a cold wall. Trettel and Larsson [24] further proposed a corrected form, but this formula is not in analytic form and cannot be used to construct the wall function method. Therefore, an analytic viscous sublayer velocity law of the wall needs to be derived theoretically. As for the logarithmic layer, the White compressible velocity law of the wall [10] based on the Walz relation is generally adopted. However, as mentioned previously, the Walz relation needs to be replaced by Zhang's new relation to involve the cold wall effect, and the temperature–velocity relation needs to be further extended to the enthalpy–velocity relation to include the chemical reaction effect. Therefore, the velocity law of the wall in the logarithmic layer also needs to be rederived. From the previously given analysis, it can be found that a series of detailed works in various aspects need to be carried out to construct the wall function method for HRTBLs.

This paper aims to develop an improved wall function method for RANS simulation of HRTBLs by theoretically deriving the generalized enthalpy–velocity relation and velocity law of the wall, which involve the chemical reaction and severe cold wall effects. The paper is organized as follows. The governing equations and numerical methods are presented in Sec. II. The generalized enthalpy–velocity relation and velocity law of the wall are theoretically derived and verified by numerical simulations in Sec. III. In Sec. IV, an improved wall function method for HRTBLs is designed based on the new analytic laws of the wall, and its coupling strategy with CFD code is further established. After that, the application studies of the improved wall function method are carried out for typical HRTBLs over the flat plate. The accuracy of c_f and q_w predicted by the improved wall function method and the efficiency improvement in numerical simulations are analyzed and discussed. Finally, the conclusions are presented in Sec. V.

II. Numerical Methods and Validation

A. Governing Equations and Numerical Methods

The studies in this paper focus on HRTBLs with a high total enthalpy level of freestream, and there will be intense dissociation/recombination chemical reactions of high-temperature air in the flow field, which means the chemical reaction effect needs to be considered. The thermal nonequilibrium effect is assumed to be insignificant, and the one-temperature model will be used in present studies. Therefore, the Favrè-averaged Navier–Stokes equations including chemical reactions are used, which can be written in a conservation form under the Cartesian coordinate system as

$$
\begin{aligned}
&\frac{\partial(\bar{\rho}\tilde{Y}_s)}{\partial t} + \frac{\partial}{\partial x_j}(\bar{\rho}\tilde{Y}_s\tilde{u}_j) = \frac{\partial}{\partial x_j}\left[\left(\bar{\rho}D_s + \frac{\mu_t}{Sc_t}\right)\frac{\partial \tilde{Y}_s}{\partial x_j}\right] + \bar{\dot{\omega}}_s, \\
&\qquad (s = 1, 2, \ldots, ns) \\
&\frac{\partial \bar{\rho}}{\partial t} + \frac{\partial}{\partial x_j}(\bar{\rho}\tilde{u}_j) = 0 \\
&\frac{\partial(\bar{\rho}\tilde{u}_i)}{\partial t} + \frac{\partial}{\partial x_j}(\bar{\rho}\tilde{u}_i\tilde{u}_j + \delta_{ij}\bar{p}) = \frac{\partial}{\partial x_j}(\bar{\tau}_{ij} + \tau^t_{ij}) \\
&\frac{\partial(\bar{\rho}\tilde{E})}{\partial t} + \frac{\partial}{\partial x_j}(\bar{\rho}\tilde{H}\tilde{u}_j) = \frac{\partial}{\partial x_j}[\tilde{u}_i(\bar{\tau}_{ij} + \tau^t_{ij})] \\
&\qquad + \frac{\partial}{\partial x_j}\left(\lambda\frac{\partial \tilde{T}}{\partial x_j} + \frac{\mu_t}{Pr_t}\frac{\partial \tilde{h}}{\partial x_j} + \sum_{s=1}^{ns}\bar{\rho}D_s\tilde{h}_s\frac{\partial \tilde{Y}_s}{\partial x_j}\right)
\end{aligned}
\tag{1}
$$

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The equation of state for the gas mixture is given as

$$\bar{p} = \bar{\rho}\,\tilde{T}\,R_u \sum_{s=1}^{ns} \frac{\tilde{Y}_s}{M_s} \tag{2}$$

where $\bar{\rho}$, $\tilde{u}_j$, $\bar{p}$, and $\tilde{T}$ are the density, velocities, pressure, and temperature, respectively. M_s, $\tilde{Y}_s$, D_s, and $\tilde{h}_s$ are the molecular weight, mass fraction, mass diffusion coefficients, and static enthalpy (the sum of the sensible and formation enthalpy) of species s, and $\bar{\dot{\omega}}$ is the mass production per unit volume per unit time of species s. $\tilde{H}$ is the total enthalpy. $\tilde{E}$ is the total energy per unit mass of the gas mixture:

$$\tilde{E} = \sum_{s=1}^{ns} \tilde{Y}_s \tilde{h}_s + \frac{1}{2}\tilde{u}_j\tilde{u}_j - \frac{\bar{p}}{\bar{\rho}} \tag{3}$$

$\bar{\tau}_{ij}$ and τ^t_{ij} are the molecular stress tensor and Reynolds stress tensor, respectively:

$$\bar{\tau}_{ij} = \mu\left(\frac{\partial\tilde{u}_i}{\partial x_j} + \frac{\partial\tilde{u}_j}{\partial x_i}\right) - \frac{2}{3}\mu\frac{\partial\tilde{u}_k}{\partial x_k}\delta_{ij}$$
$$\tau^t_{ij} = \mu_t\left(\frac{\partial\tilde{u}_i}{\partial x_j} + \frac{\partial\tilde{u}_j}{\partial x_i}\right) - \frac{2}{3}\mu_t\frac{\partial\tilde{u}_k}{\partial x_k}\delta_{ij} \tag{4}$$

where μ and μ_t are the molecular and eddy viscosity, respectively. In this paper, μ_t is calculated by the Spalart–Allmaras turbulence model [25]. The molecular viscosity, thermal conductivity, and mass diffusion coefficients for each species are obtained by the fitting polynomial proposed by Gupta et al., whose fitting coefficients can be obtained from [26]. Then, the transport coefficients for the gas mixture are calculated through Wilke's mixing law [27]. The specific heat capability and static enthalpy of all the species are fitted by temperature polynomials, whose fitting coefficients can be accessed from [28].

The chemical reaction mass production term $\bar{\dot{\omega}}_s$ in Eq. (1) represents the effect of chemical reaction on the mass fraction of each species. In this paper, the chemical kinetic model with five-species (N_2, O_2, NO, N, O) and six-reactions Eq. (5) proposed by Gupta et al. [26] is adopted to simulate the chemical reactions of high-temperature air:

$$\begin{aligned}
&N_2 + M_1 \rightleftharpoons 2N + M_1\\
&O_2 + M_2 \rightleftharpoons 2O + M_2\\
&NO + M_3 \rightleftharpoons N + O + M_3\\
&N_2 + O \rightleftharpoons NO + N\\
&NO + O \rightleftharpoons O_2 + N\\
&N_2 + N \rightleftharpoons 3N
\end{aligned} \tag{5}$$

where M_1 (O, O_2, N_2, NO), M_2 (O, N, O_2, N_2, NO), and M_3 (O, N, O_2, N_2, NO) are the third body of the chemical reactions [26].

The governing Eqs. (1) will be solved using ACANS, a finite difference CFD code developed by the authors. The reliability of ACANS has been verified by a variety of numerical simulation studies [1,29,30]. In the present numerical simulations, the Roe scheme [31] and second-order central difference scheme are employed to discretize the inviscid and viscous flux, respectively. To enlarge the value of the Courant–Friedrichs–Lewy (CFL) number and improve the stability of numerical simulations, the implicit lower–upper symmetric Gauss–Seidel method [32] is employed for the time marching, and the local timestep is adopted to accelerate the convergence.

B. Numerical Validation

In this section, two wind tunnel experimental cases are selected to validate the reliability of the adopted numerical methods. The first case is a hypersonic turbulent boundary layer over the flat plate, which will be carried out with a dense mesh (without wall function method) to examine the reliability of the numerical methods for RANS simulation of HTBLs. Another case is a hypersonic chemical nonequilibrium laminar flow around a cylinder, which will be used to examine the simulation accuracy of chemical nonequilibrium flows.

1. *Hypersonic Turbulent Boundary Layer Flow over the Flat Plate*

Holden's [33] wind tunnel experiment on the hypersonic turbulent boundary layer flow over the flat plate is first simulated. The computational domain is shown in Fig. 1, which is 1 m in the streamwise direction and 0.8 m in the normal direction, with a two-dimensional orthogonal structured mesh (100×120). To verify the grid independence, the coarse, medium, and dense meshes with $\Delta y_1 = 3 \times 10^{-5}$ m, $\Delta y_1 = 5 \times 10^{-6}$ m, and $\Delta y_1 = 3 \times 10^{-6}$ m (Δy_1^+ are 3, 0.5, and 0.3, respectively) are generated. The boundary conditions for the numerical simulation are collected from [33], as shown in Table 1, where the wall temperature ratio Θ is defined as the ratio of the wall temperature T_w to adiabatic wall temperature $T_r = T_e + ru_e^2/(2c_p)$. $\Theta < 1$ implies a cold wall boundary, and the cold wall effect will be more significant with the decrease of Θ. Moreover, the total temperature of this free-stream condition is just around 1500 K, so there is no need to consider the high-temperature chemical reactions. Hence, this numerical simulation will be carried out based on the calorically prefect-gas assumption.

Figures 2a and 2b show the comparisons of skin friction c_f and Stanton number St between numerical results and the experimental data [33], and the theoretical results of Van Driest II [12] are employed for reference. It can be seen that the numerical results agree fairly well with the experimental data and Van Driest II theoretical results, indicating that the Spalart–Allmaras turbulence model adopted in this paper can correctly simulate HTBLs. In addition, the result obtained by the medium mesh is almost identical to that of the dense mesh, indicating that the medium and dense meshes are enough to achieve grid independence.

2. *Hypersonic Chemical Nonequilibrium Laminar Flow Around a Cylinder*

To further validate the accuracy of the numerical methods in simulating chemical nonequilibrium flows, Hannemann et al.'s experiment on hypersonic laminar reacting flow around a cylinder [34] will be numerically simulated. The diameter of the experimental cylinder is $D = 90$ mm, and the computational mesh is shown in Fig. 3. Three meshes with $\Delta y_1 = 5 \times 10^{-7}$ m, $\Delta y_1 = 1 \times 10^{-7}$ m, and $\Delta y_1 = 5 \times 10^{-8}$ m are generated to verify the grid independence. The boundary

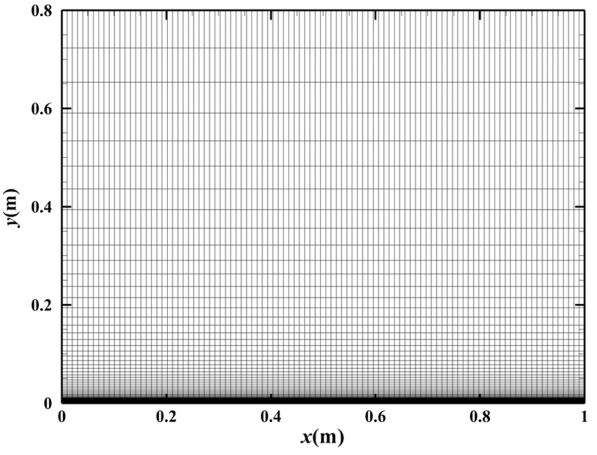


Fig. 1 Two-dimensional orthogonal structural mesh for flat plate boundary layer case.

Table 1 Boundary conditions for numerical simulation

Ma_∞	T_∞, K	p_∞, Pa	Re, m^{-1}	T_w, K	Θ
11.1	64	1741.83	3.93×10^7	296.7	0.20

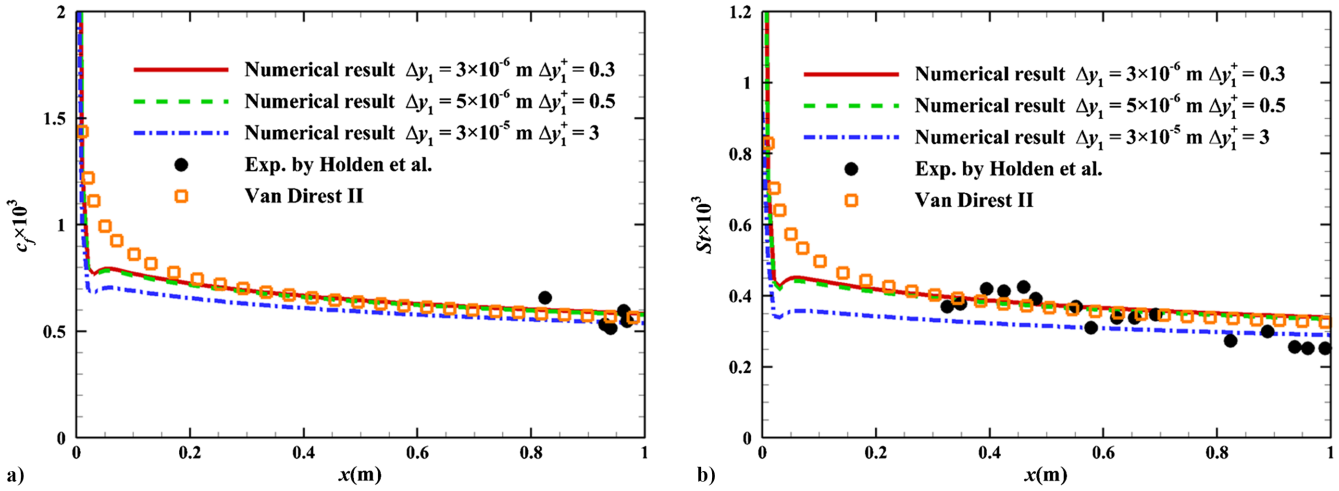


Fig. 2 Comparisons of a) c_f and b) St between numerical results and the experimental data [33]. The results of Van Driest II are employed for reference.

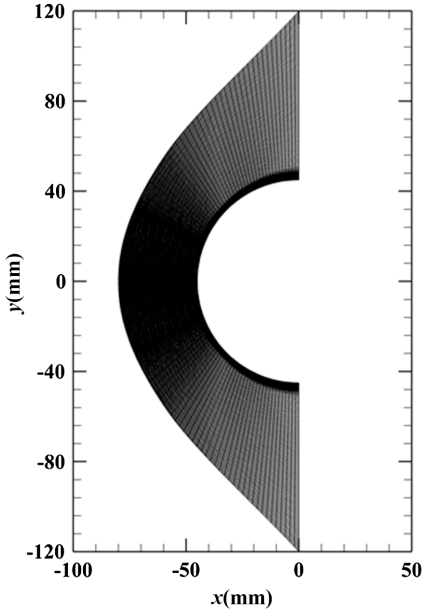


Fig. 3 Two-dimensional structural mesh for cylinder case.

conditions for this numerical case are consistent with the experimental freestream and wall conditions in Table 2. The total enthalpy calculated by the parameters in Table 2 is 11.4 MJ/kg, which means there will be intense chemical reactions in the flow field after the shock wave, and the chemical reaction model of high-temperature air needs to be considered. In this case, the wall boundary condition for species is full-catalytic:

$$\tilde{Y}_{s,w} = \tilde{Y}_{s,\infty} \tag{6}$$

The comparisons of wall pressure p_w and wall heat flux q_w between numerical results and the experimental data are given in Fig. 4, illustrating that the numerical results agree well with the experimental data. Besides, the result obtained by the medium mesh is almost identical to that of the dense mesh, indicating that the medium and dense meshes are enough to achieve grid independence. These results prove that the numerical methods and chemical kinetic model used in this paper could simulate the chemical nonequilibrium flows accurately.

Table 2 Freestream and wall conditions of Hannemann's hypersonic reacting cylinder experiment [34]

Ma_∞	T_∞, K	p_∞, Pa	T_w, K	Y_{N2}	Y_{O2}	Y_{NO}	Y_N	Y_O
8.78	694	687	300	0.735	0.134	0.051	0.0	0.080

III. Laws of the Wall for Hypersonic Reacting Turbulent Boundary Layers

The key to constructing the wall function method is the analytic form of laws of the wall for turbulent boundary layers, especially accurate laws in the near-wall region. Therefore, in order to design an improved wall function method for HRTBLs, the generalized enthalpy–velocity relation and velocity law of the wall involving the chemical reaction and severe cold wall effects are theoretically derived in this section, and their accuracy is examined by numerical simulations of typical HRTBLs.

A. Generalized Enthalpy–Velocity Relation

For the zero-pressure gradient compressible turbulent boundary layers, Walz obtained a temperature–velocity relation based on the Crocco–Busemann temperature–velocity relation [12] by introducing the recovery factor r to involve the $Pr \neq 1$ effect of air. This relation is known as the Walz relation:

$$\tilde{T} = \tilde{T}_w + (\tilde{T}_r - \tilde{T}_w)\frac{\tilde{u}}{\tilde{u}_e} + (\tilde{T}_e - \tilde{T}_r)\left(\frac{\tilde{u}}{\tilde{u}_e}\right)^2 \tag{7}$$

However, the previous DNS studies by Zhang et al. [17] and Duan et al. [19] indicate that Eq. (7) is only valid under the adiabatic or quasi-adiabatic wall conditions while showing significant deviation from the DNS data under severe cold wall conditions. To overcome this defect, Zhang et al. [17] introduced a generalized recovery factor r_g to consider $Pr \neq 1$ and cold wall effects comprehensively and thereafter obtained a new temperature–velocity relation for compressible turbulent boundary layers:

$$\tilde{T} = \tilde{T}_w + (\tilde{T}_{rg} - \tilde{T}_w)\frac{\tilde{u}}{\tilde{u}_e} + (\tilde{T}_e - \tilde{T}_{rg})\left(\frac{\tilde{u}}{\tilde{u}_e}\right)^2 \tag{8}$$

where $\tilde{T}_{rg} = \tilde{T}_e + r_g\tilde{u}_e^2/(2c_p)$ is the local generalized recovery temperature, and r_g is defined as

$$r_g = \frac{\tilde{T}_w - \tilde{T}_e}{\tilde{u}_e^2/(2c_p)} + \frac{2\,Pr}{\tilde{u}_e}\frac{q_w}{\tau_w} \tag{9}$$

Subsequently, according to the DNS studies by Zhang et al. [17], Mo et al. [1], and Zhang et al. [2], under the calorically prefect-gas assumption, Eq. (8) is able to accurately consider the severe cold wall effect on temperature–velocity distribution in HTBLs.

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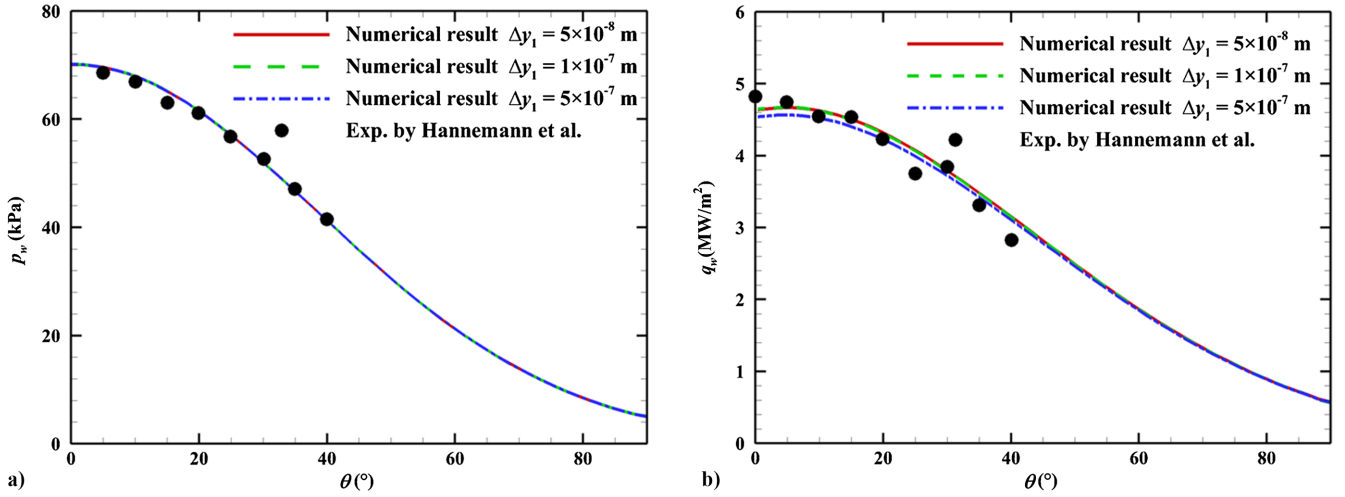


Fig. 4 Comparisons of a) p_w and b) q_w between numerical results and the experimental data [34].

As mentioned in Sec. I, it is no longer appropriate to use the temperature–velocity relation in HRTBLs due to the fact that high-temperature air chemical reactions will significantly change the temperature distribution in HRTBLs. To illustrate this point, we consider two sharp wedges with half wedge angles α of 15° and 20° flying at a Mach number of 25 and an altitude of 25 km, as shown in Fig. 5a. The postshock flow parameters are chosen as typical freestream conditions for numerical experiments of HRTBLs on the flat plate. The previously given typical hypersonic freestream and wall conditions are collected in Table 3 and named Wedge15H25 and Wedge20H25, respectively, where the notation Wedge15H25, for example, denotes that $\alpha = 15°$ and the flight altitude is 25 km. Both cases have a total enthalpy of 28 MJ/kg, leading to intense chemical reactions of high-temperature air in turbulent boundary layers. Moreover, the wall temperature ratio Θ is 0.1, indicating that there will be a severe cold wall effect in HRTBLs as well. Figure 5b shows the numerical subdomain for the RANS simulation of HRTBLs, which employs the two-dimensional orthogonal structured mesh (100 × 120) with 1 m in the streamwise direction and 0.8 m in the normal direction. The streamwise grids are distributed uniformly, and the wall-normal grids are refined by geometric distribution function with $\Delta y_1 = 5 \times 10^{-7}$ m and $\Delta y_1^+ < 0.4$. Grid resolution studies in the appendix show that grid independence is achieved on this mesh. The left side of the computational domain adopts the far-field boundary condition, whereas the upper and right sides adopt the outflow boundary condition. The isothermal and non-catalytic wall boundary condition [20] is adopted for the wall.

Figures 6a and 6b provide comparisons of the temperature–velocity distribution among the numerical results, the Walz relation Eq. (7), and the Zhang's relation Eq. (8) in HRTBLs of Wedge15H25 and Wedge20H25, respectively. It can be seen that under the chemical reaction and severe cold wall conditions, the traditional Walz relation Eq. (7) has completely failed with significant deviation from the numerical results, which is consistent with the conclusion of Duan and Martin's DNS study [20]. The Zhang's relation Eq. (8) is much closer to the numerical results, illustrating that it considers the cold wall effect. However, the real temperature distributions are significantly lower than Eq. (8) due to the heat-absorbing dissociation chemical reactions of high-temperature air occurring in HRTBLs. Furthermore, for the two freestream conditions shown in Table 3, there will be more significant dissociation reactions in the latter than in the former, and it can be seen from the comparison between Figs. 6a and 6b that the deviation of Eq. (8) will become more obvious as the chemical reaction effect enhances. To illustrate this conclusion more clearly, Table 4 further provides the relative errors between Eq. (8) and numerical temperature–velocity distributions at the typical positions in HRTBLs, where the relative errors of the latter are larger than those of the former.

Table 3 Freestream and wall parameters for the RANS simulation of HRTBLs in subdomain

Case	Ma_e	T_e, K	p_e, kPa	Re, m^{-1}	T_w, K	Θ
Wedge15H25	7.4	2473	183.7	2.39×10^7	2600	0.1
Wedge20H25	6.0	3469	302.5	2.09×10^7	2600	0.1

To remove the explicit dependence of the temperature–velocity relation on chemical models, Duan et al. [20] introduced the concept of nondimensional 'recovery enthalpy' and proposed the following semiempirical enthalpy–velocity relation:

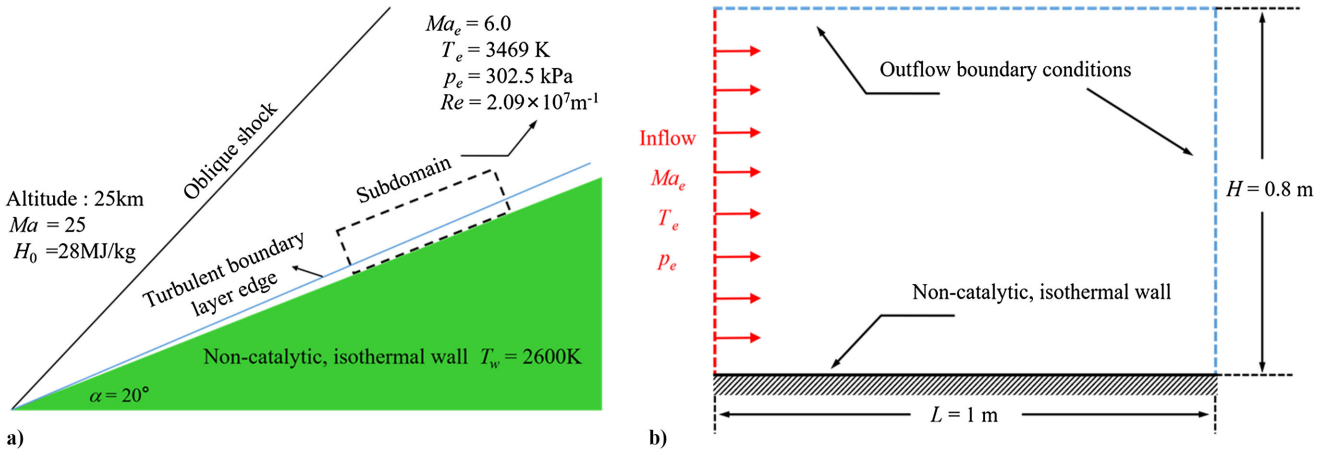


Fig. 5 a) Physical model and b) numerical subdomain for the RANS simulation of HRTBLs.

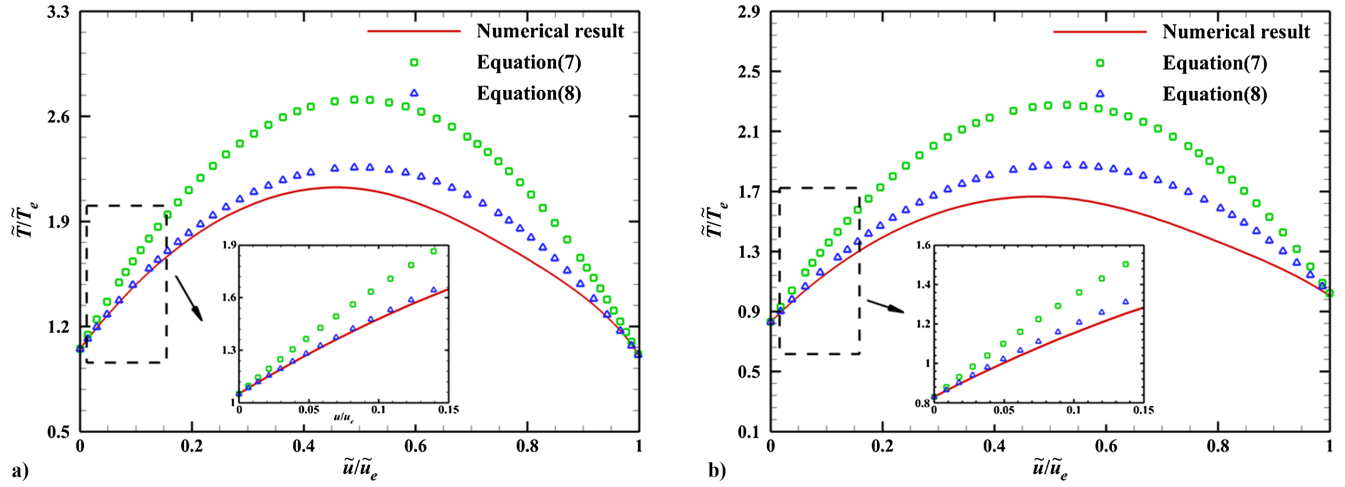


Fig. 6 Comparisons of temperature–velocity distributions among numerical results, Eq. (7), and Eq. (8) in HRTBLs of a) Wedge15H25 and b) Wedge20H25.

Table 4 Relative errors between Eq. (8) and numerical temperature–velocity distributions at typical positions in HRTBLs

Case	$\tilde{u}/\tilde{u}_e$	$\tilde{T}/\tilde{T}_e$ (Numerical)	$\tilde{T}/\tilde{T}_e$ [Eq. (8)]	Errors of Eq. (8)
Wedge15H25	0.1	1.48	1.50	1.35%
	0.2	1.79	1.84	2.79%
	0.3	2.00	2.08	4.00%
Wedge20H25	0.1	1.15	1.19	3.48%
	0.2	1.39	1.48	6.47%
	0.3	1.56	1.69	8.33%

$$\tilde{h} = \tilde{h}_w + (\tilde{h}_r - \tilde{h}_w) f\left(\frac{\tilde{u}}{\tilde{u}_e}\right) - \frac{r\tilde{u}^2}{2} \tag{10}$$

where

$$f\left(\frac{\tilde{u}}{\tilde{u}_e}\right) = 0.8259\left(\frac{\tilde{u}}{\tilde{u}_e}\right) + 0.1741\left(\frac{\tilde{u}}{\tilde{u}_e}\right)^2 \tag{11}$$

is given by the fitting of the DNS data of HRTBLs [20]. The accuracy of the Duan's semiempirical enthalpy–velocity relation Eq. (10) has been validated by many DNS studies on HRTBLs [20,21,35]. In addition, considering that the static enthalpy is a sum of the sensible enthalpy and formation enthalpy and is not explicitly affected by chemical reactions, Gao et al. [22] first proposed to extend the Walz temperature–velocity relation to a relation between static enthalpy and velocity to involve the chemical reaction effect in their study on the wall function method for supersonic combustion turbulent boundary layers. Inspired by their works, a similar approach is employed in this paper to extend the Zhang's relation Eq. (8) to the following generalized enthalpy–velocity relation:

$$\tilde{h} = \tilde{h}_w + (\tilde{h}_{rg} - \tilde{h}_w)\frac{\tilde{u}}{\tilde{u}_e} + (\tilde{h}_e - \tilde{h}_{rg})\left(\frac{\tilde{u}}{\tilde{u}_e}\right)^2 \tag{12}$$

where $\tilde{h}_{rg} = \tilde{h}_e + r_g^*\tilde{u}_e^2/2$ is the local generalized recovery enthalpy, and the new generalized recovery factor r_g^* is defined as

$$r_g^* = \frac{\tilde{h}_w - \tilde{h}_e}{\tilde{u}_e^2/2} + \frac{2Pr_w}{\tilde{u}_e}\frac{q_w}{\tau_w} \tag{13}$$

To investigate the accuracy of Eq. (12) in HRTBLs, the comparisons of enthalpy–velocity distributions among the numerical result, Duan's semiempirical enthalpy–velocity relation Eq. (10), and the generalized enthalpy–velocity relation Eq. (12) in HRTBLs for the cases of Wedge15H25 and Wedge20H25 are shown in Figs. 7a and 7b, respectively. Compared with the results in Fig. 6, it can be found that the theoretical prediction of Eq. (12) is significantly better in agreement with the numerical results. In particular, Eq. (12) almost completely collapses with the RANS simulation results in the near-wall region

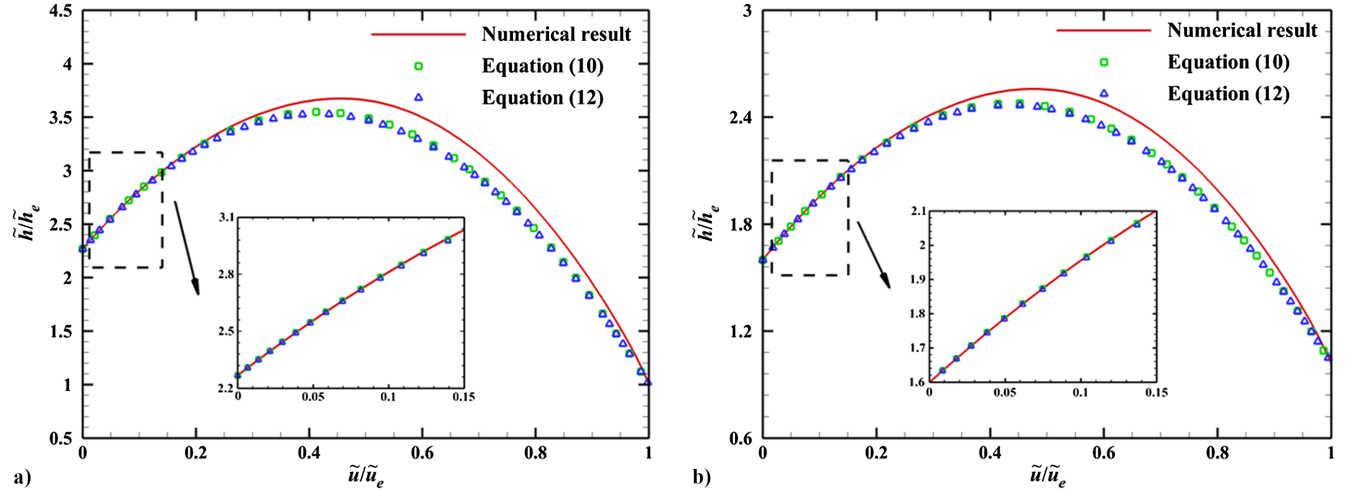


Fig. 7 Comparisons of enthalpy–velocity distributions among numerical results, Eq. (10), and Eq. (12) in HRTBLs of a) Wedge15H25 and b) Wedge20H25.

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(both are enlarged and shown in Figs. 7a and 7b), which is greatly beneficial for the subsequent design of the improved wall function method. Besides, it can be seen that Eq. (12) collapses very well with Eq. (10), with a relative error less than 0.5%. These results indicate that the chemical reaction and severe cold wall effects in HRTBLs could be comprehensively considered by extending Zhang's relation to a generalized enthalpy–velocity relation.

Finally, to facilitate the design of the improved wall function method in Sec. IV, the $\tilde{u}$ on the right side of Eq. (12) is transformed to u^+, and then Eq. (12) can be rewritten as

$$\tilde{h} = \tilde{h}_w[1 + \beta^* u^+ - \Gamma^*(u^+)^2] \tag{14}$$

where β^* and Γ^* are respectively defined as

$$\beta^* = \frac{u_\tau\left(\tilde{h}_{rg} - \tilde{h}_w\right)}{\tilde{u}_e \tilde{h}_w}, \Gamma^* = \frac{r_g^* u_\tau^2}{2\tilde{h}_w} \tag{15}$$

B. Velocity Law of the Wall for Hypersonic Reacting Turbulent Boundary Layers

The generalized enthalpy–velocity relation is theoretically proposed in the previous section. Then, the velocity law of the wall for HRTBLs will be derived in this section. As mentioned in Sec. I, the Nichols velocity law of the wall, which is used in the existing compressible wall functions, fails in HRTBLs. Hence, we will first analyze the reason for the failure of the Nichols velocity law of the wall in HRTBLs and then rederive more accurate analytic forms for the viscous sublayer and logarithmic layer, respectively. Finally, a unified velocity law of the wall is constructed for the inner layer of HRTBLs.

1. *Nichols Velocity Law of the Wall*

For the zero-pressure gradient incompressible turbulent boundary layers, the traditional velocity law of the wall is established for the viscous sublayer and logarithmic layer, respectively, whereas there is no exact velocity law of the wall for the buffer layer. Therefore, it is necessary to determine where the first grid points off the wall are located, and it is also essential to avoid them falling into the buffer layer when using the wall function method. These necessary operations will cause plenty of inconvenience to practical applications. To deal with this issue, Nichols and Nelson [9] introduced the unified velocity law of the wall for incompressible turbulent boundary layers derived by Spalding [36]

$$y^+ = u^+ + e^{-\kappa B}e^{\kappa u^+} - e^{-\kappa B}\left[1 + \kappa u^+ + \frac{(\kappa u^+)^2}{2} + \frac{(\kappa u^+)^3}{6}\right] \tag{16}$$

where

$$u_\tau = \sqrt{\frac{\tau_w}{\bar{\rho}_w}}, \quad u^+ = \frac{\tilde{u}}{u_\tau}, \quad y^+ = \frac{\bar{\rho}_w u_\tau y}{\mu_w} \tag{17}$$

The third term on the right side of Eq. (16) is the correction term introduced by Spalding, and the terms in the brackets are obtained by ignoring the fourth- and higher-order quantities of u^+ in the Taylor expansion of $e^{\kappa u^+}$. In the viscous sublayer, the sum of the correction term and $e^{-\kappa B}e^{\kappa u^+}$ is a small quantity compared with u^+. At this point, Eq. (16) approximately reduces to the classical incompressible viscous sublayer velocity law of the wall $y^+ = u^+$. In the logarithmic layer, the correction term tends to be u^+, and the right side of Eq. (16) tends to be $e^{-\kappa B}e^{\kappa u^+}$, which means that Eq. (16) reduces to the classical incompressible logarithmic layer velocity law of the wall $u^+ = 1/\kappa \ln y^+ + B$. Therefore, the wall function method does not need to determine where the first grid points off the wall are located when using Eq. (16) as the velocity law of the wall. Accordingly, Eq. (16) is called the unified velocity law of the wall for the inner layer of incompressible turbulent boundary layers. Based on Eq. (16), Nichols introduced the White compressible velocity law of the wall [10]

$$\begin{aligned} y^+_{\text{white}} &= e^{\left((\kappa/\sqrt{\Gamma})\left\{\sin^{-1}\left[\frac{(2\Gamma u^+ - \beta)}{Q}\right] - \varphi\right\}\right)} \times e^{-\kappa B} \\ \Gamma &= \frac{r u_\tau^2}{2c_p \tilde{T}_w}, \quad \beta = \frac{q_w \mu_w}{\bar{\rho}_w \tilde{T}_w \lambda_w u_\tau}, \quad \varphi = \sin^{-1}\left(\frac{-\beta}{Q}\right), \\ Q &= (\beta^2 + 4\Gamma)^{1/2} \end{aligned} \tag{18}$$

to consider the compressibility and heat transfer effects in high-speed flows. Substituting Eq. (18) into Eq. (16) to replace the term $e^{-\kappa B}e^{\kappa u^+}$ yields the Nichols velocity law of the wall

$$y^+ = u^+ + y^+_{\text{white}} - e^{-\kappa B}\left[1 + \kappa u^+ + \frac{(\kappa u^+)^2}{2} + \frac{(\kappa u^+)^3}{6}\right] \tag{19}$$

Next, the performance of the Nichols velocity law of the wall Eq. (19) will be examined based on the numerical results in Sec. III.A. Figures 8a and 8b show the comparisons of u^+ between numerical results and the Nichols velocity law of the wall Eq. (19) in HRTBLs for the cases of Wedge15H25 and Wedge20H25, respectively. It can be seen that Eq. (19) shows a significant deviation from numerical results in the viscous sublayer ($y^+ < 10$), and although there is only a slight discrepancy in the logarithmic layer, the relative error still cannot be ignored. Table 5 further provides the relative errors between Eq. (19) and the numerical u^+ at the typical positions of the viscous sublayer ($y^+ = 5$) and logarithmic layer ($y^+ = 200$ and 400)

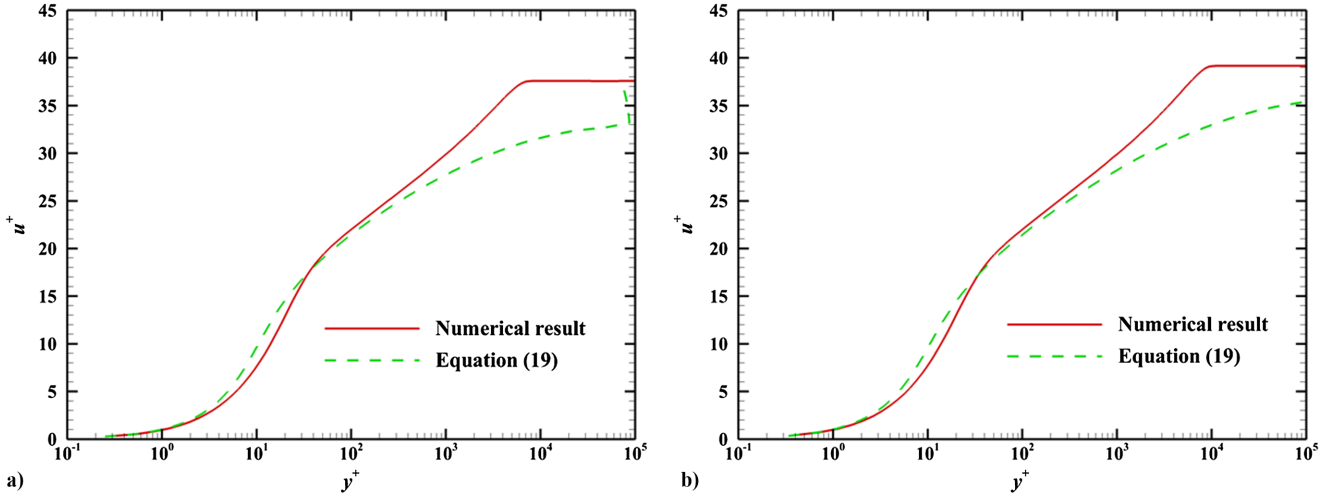


Fig. 8 Comparisons of u^+ between numerical results and Eq. (19) in HRTBLs of a) Wedge15H25 and b) Wedge20H25.

Table 5 Relative errors between Eq. (19) and the numerical u^+ at typical positions in HRTBLs of Wedge15H25 and Wedge20H25

Case	y^+	u^+ (Numerical)	u^+ [Eq. (19)]	Errors of Eq. (19)
Wedge15H25	5	4.21	5.04	19.71%
	200	24.31	23.60	2.92%
	400	26.64	25.52	4.20%
Wedge20H25	5	4.27	5.04	18.03%
	200	24.34	23.71	2.59%
	400	26.68	25.77	3.41%

in HRTBLs, where the relative errors are about 4.2% in the logarithmic layer and reach 19.7% in the viscous sublayer.

Then, the reasons for the failure of Eq. (19) in HRTBLs will be theoretically analyzed. The theoretical basis of the velocity law of the wall in the inner layers of compressible turbulent boundary layers is the constant shear stress assumption [12,37], which assumes that the total shear stress τ_{xy}, the sum of the molecular viscous shear stress $\tau = \mu\partial\tilde{u}/\partial y$, and the Reynolds shear stress, $\tau_t = \mu_t\partial\tilde{u}/\partial y$, remain essentially constant in a specific range along the wall-normal direction and approximately equal to the wall shear stress τ_w, namely,

$$\tau_{xy} = (\mu + \mu_t)\frac{\partial\tilde{u}}{\partial y} \approx \tau_w \tag{20}$$

This near-wall region with constant shear stress is generally called the *constant shear stress layer* [37,38], which essentially collapses with the inner layers of compressible turbulent boundary layers on the flat plate. In the viscous sublayer, the molecular viscous shear stress dominates, and the Reynolds shear stress is almost zero, which yields $\tau_{xy} \approx \mu\partial\tilde{u}/\partial y \approx \tau_w$. Meanwhile, in the logarithmic layer, the Reynolds shear stress is dominant, and the molecular viscous shear stress can be ignored, which yields $\tau_{xy} \approx \mu_t\partial\tilde{u}/\partial y \approx \tau_w$. Zhang et al.'s [38] and Griffin et al.'s [39] DNS studies showed that there is still a *constant shear stress layer* in HTBLs. Moreover, according to Duan and Martin's reports [20], the chemical reactions of high-temperature air do not significantly influence the distribution characteristics of the Reynolds shear stress in the *constant shear stress layer* of HRTBLs. Here, numerical distributions of the τ, τ_t, and τ_{xy} in HRTBLs are shown in Fig. 9. It can be seen that the RANS method effectively simulates the *constant shear stress layer* in the inner layers of turbulent boundary layers.

Therefore, based on the constant shear stress assumption in HRTBLs, it can be derived from Eq. (20) in the viscous sublayer to obtain

$$\mathrm{d}y^+ = \frac{\mu}{\mu_w}\mathrm{d}u^+ \Rightarrow y^+ = \int_0^{u^+}\frac{\mu}{\mu_w}\mathrm{d}u^+ = U^+ \tag{21}$$

This equation is generally called the viscosity-based velocity transformation in the studies of compressible turbulent boundary layers [23,37,40]. Moreover, it is worth noting that the Trettel–Larsson transformation [24], which has been verified by the previous DNS studies on HRTBLs [41], is equivalent to Eq. (21) in the viscous sublayer. Because the molecular viscosity of the gas mixture $\mu = \mu(\tilde{T}, \tilde{Y}_s)$ is variant with the mass fraction of species and temperature, we can obtain $\mu \approx \mu_w$ based on the calorically prefect-gas assumption when the temperature remains constant in the viscous sublayer (incompressible flows or adiabatic wall). After that, a linear velocity law of the wall $y^+ = u^+$ can be obtained directly from Eq. (21). However, the variation of molecular viscosity should be considered in the viscous sublayer of HRTBLs due to the chemical reaction and severe cold wall effects. At this point, the relation between y^+ and u^+ is nonlinear. Therefore, the failure of Eq. (19) in the viscous sublayer is caused by inconsequently ignoring the variation of molecular viscosity. Here, if the right-hand side of Eq. (21) is denoted as U^+, there will be a linear relation $y^+ = U^+$, which is similar to the relation in incompressible flows. Figure 10 shows the distributions of $u^+ - y^+$ and $U^+ - y^+$ in the viscous sublayer of HRTBLs, and the incompressible theoretical relation $u^+ = y^+$ is employed as a reference. It can be obviously seen that the distribution of $u^+ - y^+$ significantly deviates from $u^+ = y^+$ due to the chemical reaction and severe cold wall effects, whereas the $U^+ - y^+$ distribution given by Eq. (21) collapses well with $u^+ - y^+$, indicating that Eq. (21) achieves a universal scaling for the mean velocity in the viscous sublayer under different chemical reaction and cold wall conditions.

On the other hand, in the logarithmic layer where the Reynolds shear stress dominates, the following Van Driest transformation [42] could be obtained:

$$\frac{\mathrm{d}u^+}{\mathrm{d}y^+} = \frac{1}{\kappa y^+}\sqrt{\frac{\bar{\rho}_w}{\bar{\rho}}} \tag{22}$$

according to the constant shear stress assumption and Prandtl's mixing-length model [37]. Furthermore, based on the calorically prefect-gas assumption, we assume a zero-pressure gradient across the boundary layer and then obtain

$$\frac{\bar{\rho}_w}{\bar{\rho}} = \frac{\bar{p}_w}{R\tilde{T}_w}\cdot\frac{R\tilde{T}}{\bar{p}} = \frac{\tilde{T}}{\tilde{T}_w} \tag{23}$$

Substituting the Walz relation Eq. (7) into Eq. (23) and integrating yields the White compressible velocity law of the wall Eq. (18). However, for HRTBLs, Zhang's relation needs to be introduced to replace the Walz relation to involve the cold wall effect, and the generalized enthalpy–velocity relation Eq. (12) should be further

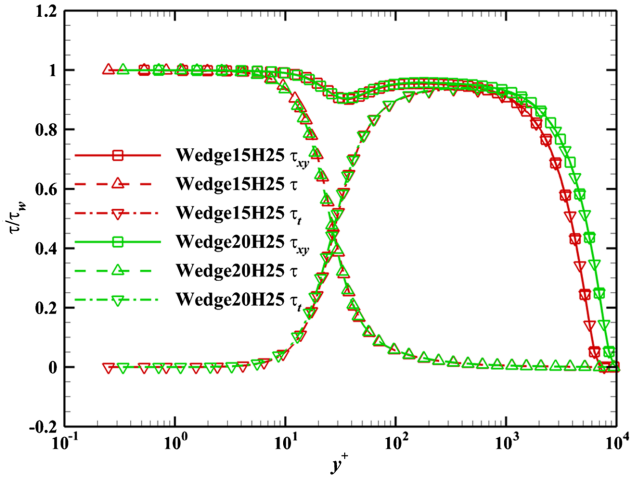


Fig. 9 Distributions of the molecular viscous shear stress τ, Reynolds shear stress τ_t, and total shear stress τ_{xy} in HRTBLs of Wedge15H25 and Wedge20H25.

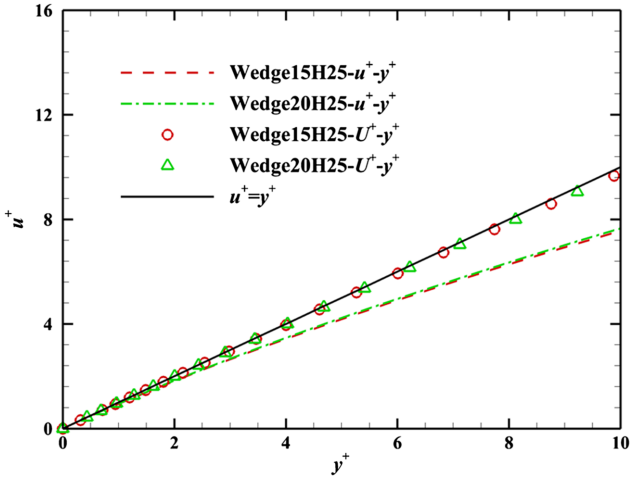


Fig. 10 Distributions of $u^+ - y^+$ and $U^+ - y^+$ in the viscous sublayer of HRTBLs. The incompressible theoretical relation $u^+ = y^+$ is employed as a reference.

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brought in to involve the chemical reaction effect. Thus, although Eq. (19) does not show significant deviation from the numerical results in the logarithmic layer, it is still necessary to use the new enthalpy–velocity relation to rederive the velocity law of the wall.

2. *New Analytic Velocity Law of the Wall in the Viscous Sublayer*

The viscosity-based velocity transformation Eq. (21) achieves a universal scaling for the mean velocity in the viscous sublayer under different chemical reaction and cold wall conditions. However, this velocity law of the wall is in integral form and cannot be used directly to construct the wall function method. Therefore, it is necessary to derive its analytic form.

The key to obtaining the analytic form of Eq. (21) is to find an approximate treatment for the variation of molecular viscosity μ along the wall-normal direction. In the studies of compressible boundary layers, an effective approach is to introduce the viscosity power law $\mu/\mu_w = (\tilde{T}/\tilde{T}_w)^{\omega}$ and the temperature–velocity relation [37,40,43]. Smits and Dussuage [40] integrated Eq. (21) by assuming $\omega = 1$ and obtained an analytic velocity law of the wall in the viscous sublayer of supersonic turbulent boundary layers. Inspired by their studies, the relation between viscosity and enthalpy in the viscous sublayer of HRTBLs will be first established, and then the analytic expression of Eq. (21) will be derived based on the new enthalpy–velocity relation Eq. (12).

For hypersonic flows, the static enthalpy of the gas mixture can be expressed as $\tilde{h} = \sum_{s=1}^{ns} \tilde{Y}_s \tilde{h}_s$. Differentiating both sides of this equation along the wall-normal direction yields

$$\frac{\mathrm{d}\tilde{h}}{\mathrm{d}y} = \sum_{s=1}^{ns} \tilde{h}_s \frac{\mathrm{d}\tilde{Y}_s}{\mathrm{d}y} + \sum_{s=1}^{ns} \tilde{Y}_s c_{ps} \frac{\mathrm{d}\tilde{T}}{\mathrm{d}y} \tag{24}$$

Equation (24) implies that the wall-normal gradient of the static enthalpy is related to the gradients of temperature and species mass fraction. Figures 11a and 11b show the profiles of the mass fraction of each species in HRTBLs for the case of Wedge15H25 and Wedge20H25. It can be found that under a noncatalytic wall boundary condition, the species mass fraction is essentially constant in the viscous sublayer and only appears to have slight variation even in the range of $y^+ < 200$. Therefore, it can be assumed that d $\tilde{Y}_s/dy = 0$ in the near-wall region (especially in the viscous sublayer), and the wall-normal variation of the static enthalpy is mainly related to the temperature gradient. Similarly, the wall-normal variation of molecular viscosity μ is also primarily caused by the temperature gradient. Hence, a new viscosity-enthalpy power law can be established as

$$\frac{\mu}{\mu_w} = \left(\frac{\tilde{h}}{\tilde{h}_w}\right)^{\omega^*} \tag{25}$$

Figures 12a and 12b provide the comparisons between the numerical distribution of molecular viscosity and the viscosity-enthalpy power law Eq. (25) in the viscous sublayer ($y^+ < 10$), where $\omega*$ is approximately 0.72. It can be obviously seen that the profiles predicted by Eq. (25) collapse fairly well with numerical results, indicating that Eq. (25) is accurate.

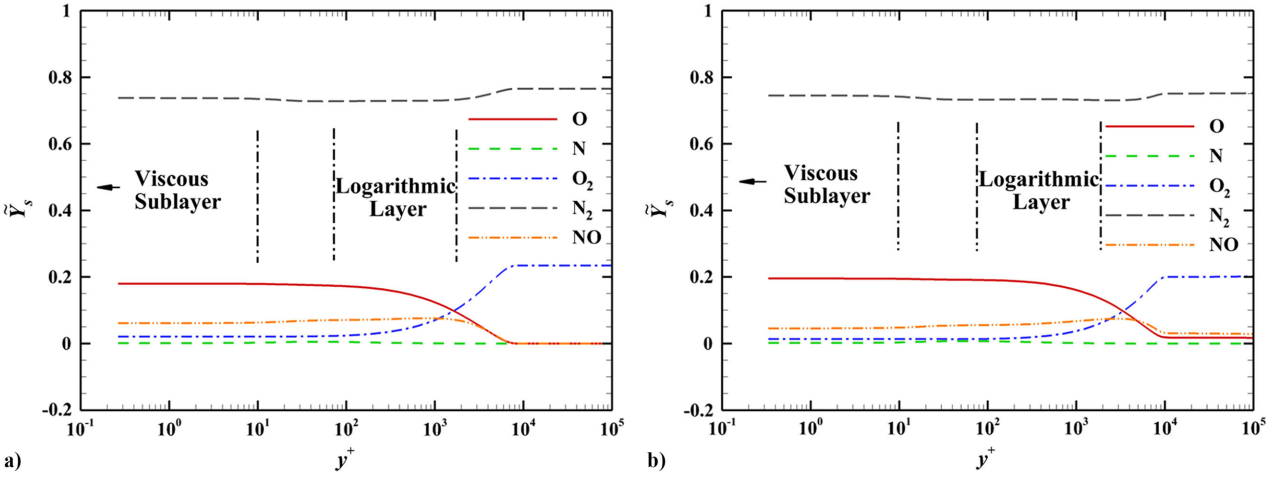


Fig. 11 Profiles of species mass fraction in HRTBLs of a) Wedge15H25 and b) Wedge20H25.

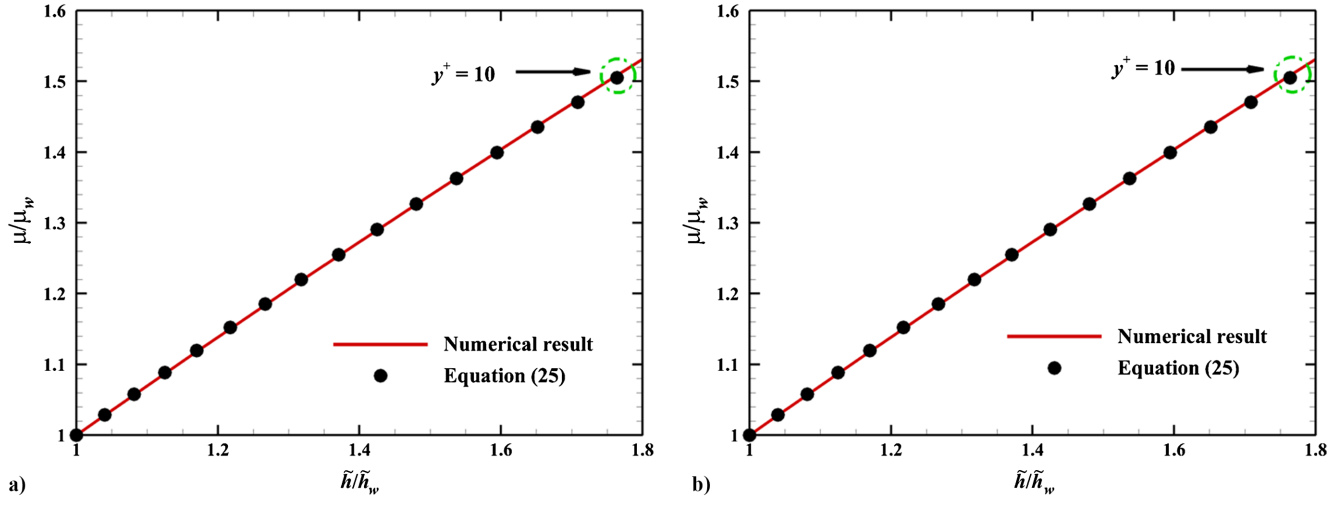


Fig. 12 Comparisons between numerical results and Eq. (25) in the viscous sublayer of HRTBLs of a) Wedge15H25 and b) Wedge20H25.

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Then, substituting Eq. (25) into Eq. (21) and introducing the enthalpy–velocity relation Eq. (12) yields

$$y^+ = \int_0^{u^+} \left[1 + \frac{\tilde{h}_{rg} - \tilde{h}_w}{\tilde{h}_w}\frac{\tilde{u}}{\tilde{u}_e} + \frac{\tilde{h}_e - \tilde{h}_{rg}}{\tilde{h}_w}\left(\frac{\tilde{u}}{\tilde{u}_e}\right)^2\right]^{\omega^*} \mathrm{d}u^+ \tag{26}$$

To transform the integral form Eq. (26) to an analytical one, the previously given integrand is subjected to Taylor expansion, assuming that $\tilde{u}/\tilde{u}_e$ is small in the viscous sublayer, and the small quantities of second and higher-order are ignored. After integration, we obtain

$$y^+ = u^+\left[1 + \omega^*\frac{\tilde{h}_{rg} - \tilde{h}_w}{2\tilde{h}_w}\frac{\tilde{u}}{\tilde{u}_e} + \omega^*\frac{\tilde{h}_e - \tilde{h}_{rg}}{3\tilde{h}_w}\left(\frac{\tilde{u}}{\tilde{u}_e}\right)^2\right] = u^+_{GM} \tag{27}$$

Eq. (27) is the new analytic velocity law of the wall for the viscous sublayer of HRTBLs proposed in this paper. Observing the expression of Eq. (27), it can be found that the terms in the brackets, except for unity, consider the chemical reaction and cold wall effects. If the right-side term of Eq. (27) is denoted as u^+_{GM}, there is also a linear relation $y^+ = u^+_{GM}$, similar to incompressible flows.

Then, the validity of analytic expression Eq. (27) will be examined based on the numerical results in Sec. III.A. Figure 13 shows the distributions of $u^+ - y^+$ and $u^+_{GM} - y^+$ in the viscous sublayer of HRTBLs of Wedge15H25 and Wedge20H25, and the incompressible theoretical relation $u^+ = y^+$ is employed as a reference. It can be observed that Eq. (27) collapses fairly well with $u^+ = y^+$ in the range of $y^+ < 5$. Although the errors gradually occur with the increase of y^+, the relative errors remain less than 5% in the range of $y^+ < 10$. Based on the previously given results, it can be concluded that the new analytic velocity law of the wall Eq. (27) is valid in the viscous sublayer of HRTBLs.

3. *Unified Form of the Velocity Law of the Wall*

Next, an analytic velocity law of the wall in the logarithmic layer of HRTBLs will be derived by using the generalized enthalpy–velocity relation Eq. (12). It can be found from the derivation of the White compressible velocity law of the wall Eq. (18) that the key is to find a new relation between the density and static enthalpy. From the equation of state for the gas mixture, the density term in Eq. (22) can be approximatively expressed as

$$\frac{\bar{\rho}_w}{\bar{\rho}} = \frac{R\tilde{T}}{R_w\tilde{T}_w} = \frac{Rc_{pw}}{R_w c_p}\cdot\frac{c_p\tilde{T}}{c_{pw}\tilde{T}_w} \approx \frac{Rc_{pw}}{R_w c_p}\cdot\frac{\tilde{h}}{\tilde{h}_w} \tag{28}$$

Figure 14 shows the distributions of the term $(Rc_{pw}/R_w c_p)$ in HRTBLs for the cases of Wedge15H25 and Wedge20H25. It can be found that this term essentially remains constant and is close to unity in the inner layer of HRTBLs. Therefore, we assume $(Rc_{pw}/R_w c_p) = 1$, and Eq. (28) can be simplified as

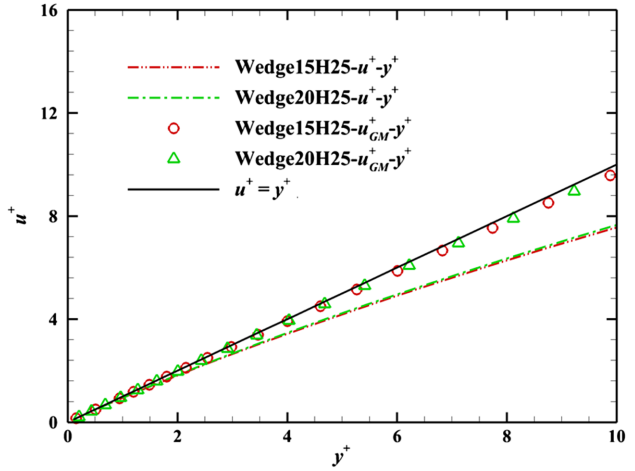


Fig. 13 Distributions of $u^+ - y^+$ and $u^+_{GM} - y^+$ in the viscous sublayer of HRTBLs. The incompressible theoretical relation $u^+ = y^+$ is employed as a reference.

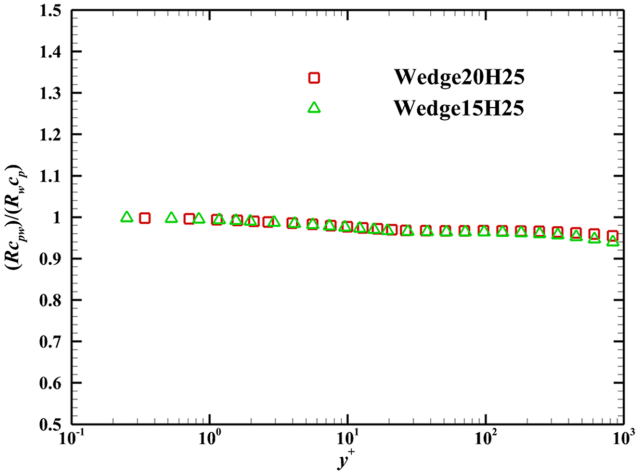


Fig. 14 Distributions of term $(Rc_{pw}/R_w c_p)$ in HRTBLs of Wedge15H25 and Wedge20H25.

$$\frac{\bar{\rho}_w}{\bar{\rho}} \approx \frac{\tilde{h}}{\tilde{h}_w} \tag{29}$$

Substituting Eq. (29) into Eq. (22) and introducing the generalized enthalpy–velocity relation Eq. (12) yields the analytic velocity law of the wall for the logarithmic layer of HRTBLs, which can include the chemical reaction and cold wall effects. After integration, we obtain

$$u^+ = (1/2\Gamma^*)\left\{\beta^* + Q^*\sin\left[\varphi^* + \frac{\sqrt{\Gamma^*}}{\kappa}\ell_n\left(\frac{y^+}{y_0^+}\right)\right]\right\}$$
$$\varphi^* = \sin^{-1}\left(\frac{-\beta^*}{Q^*}\right), \quad Q^* = [(\beta^*)^2 + 4\Gamma^*]^{1/2} \tag{30}$$

where $y_0^+ = e^{-\kappa B}$, and the definitions of β^* and Γ^* have been given in Eq. (15). Using Eq. (27) and Eq. (30) to replace the incompressible viscous sublayer term u^+ and incompressible logarithmic layer term $e^{-\kappa B}e^{\kappa u+}$ in Eq. (16), respectively, the unified form of velocity law of the wall for the inner layer of HRTBLs can be derived as follows:

$$y^+ = u^+_{GM} + \hat{y}^+_{\text{white}} - e^{-\kappa B}\left[1 + \kappa u^+ + \frac{(\kappa u^+)^2}{2} + \frac{(\kappa u^+)^3}{6}\right] \tag{31}$$

where

$$\hat{y}^+_{\text{white}} = e^{\left(\left(\kappa/\sqrt{\Gamma^*}\right)\left\{\sin^{-1}\left[\frac{(2\Gamma^* u^+ - \beta^*)}{Q^*}\right] - \varphi^*\right\}\right)} \times e^{-\kappa B} \tag{32}$$

Figures 15a and 15b show the comparisons of u^+ among the numerical results, the Nichols velocity law of the wall Eq. (19), and the new unified velocity law of the wall Eq. (31) in HRTBLs of Wedge15H25 and Wedge20H25. It can be seen that Eq. (31) collapses fairly well with numerical results. Table 6 further provides the relative errors of Eq. (19) and Eq. (31). It can be seen that compared with Eq. (19), Eq. (31) reduces the relative errors from 19.7% to less than 2.2% at the typical position of the viscous sublayer ($y^+ = 5$) and from 4.2% to less than 1.0% at the typical positions of the logarithmic layer ($y^+ = 200$ and $y^+ = 400$), showing a significant improvement over Eq. (19).

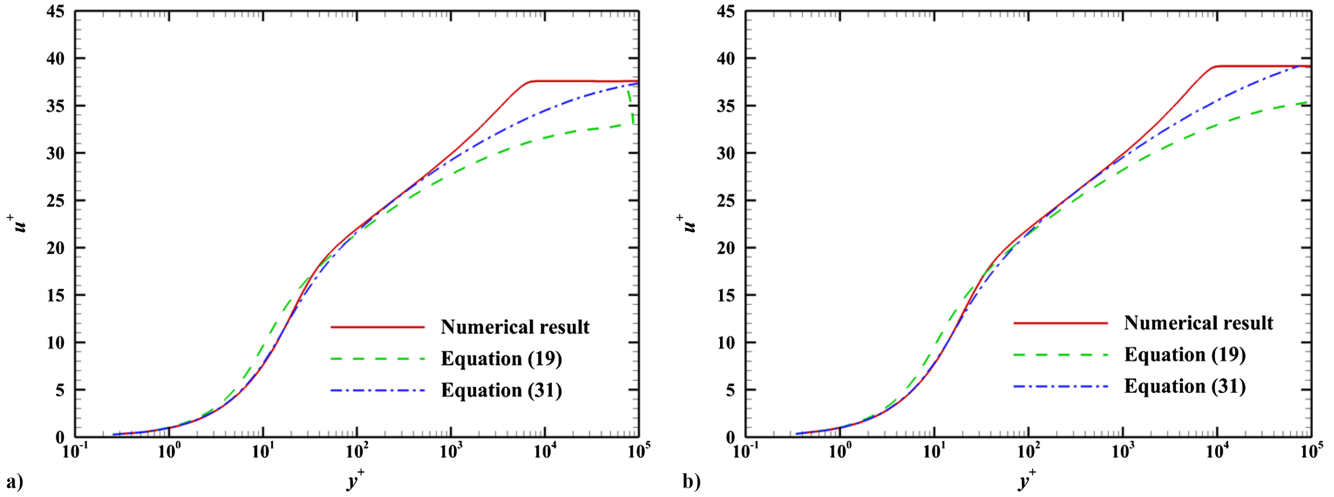


Fig. 15 Comparisons of u^+ among numerical results, Eq. (19), and Eq. (31) in HRTBLs of a) Wedge15H25 and b) Wedge20H25.

Table 6 Relative errors of Eqs. (19) and (31) at typical positions in HRTBLs

Case	y^+	Errors of Eq. (19)	Errors of Eq. (31)
Wedge15H25	5	19.71%	2.14%
	200	2.92%	0.25%
	400	4.20%	0.41%
Wedge20H25	5	18.03%	1.41%
	200	2.59%	0.35%
	400	3.41%	0.13%

IV. Design and Application of the Improved Wall Function Method

The generalized enthalpy–velocity relation and analytic velocity law of the wall involving the chemical reaction and severe cold wall effects have been derived in Sec. III. In this section, an improved wall function method for HRTBLs is designed based on the new analytic laws of the wall, and the coupling strategy between the wall function method and CFD simulation is established. Finally, the application study of the improved wall function method for typical HRTBLs over the flat plate is carried out. The accuracy of c_f and q_w predicted by the improved wall function method, as well as the efficiency improvement in numerical simulations, are analyzed and discussed.

A. Design of the Improved Wall Function Method

The wall function method needs to be coupled with CFD code as a boundary condition, and the most critical task is to replace the corresponding terms of momentum and energy equations in CFD code by wall shear stress τ_w and wall heat flux q_w obtained from the wall function method.

The coupling strategy proposed in this paper is shown in Fig. 16. Firstly, the generalized recovery factor r_g^* is calculated according to the generalized enthalpy–velocity relation Eq. (12), and then the r_g^* is further substituted into Eq. (15) to obtain β^* and Γ^*. Secondly, the u_τ and τ_w are calculated iteratively according to the velocity law of the wall Eq. (31) and the velocity at the first grid point off the wall u_1. Thirdly, the obtained τ_w is used to replace the shear stress term $\tau_{1/2}$ of the momentum equation in the CFD viscous subroutine based on the constant shear stress assumption, as shown in Fig. 16. Afterwards, according to Eq. (13) and τ_w, q_w can be expressed as

$$q_w = \left(r_g^* - \frac{\tilde{h}_w - \tilde{h}_e}{\tilde{u}_e^2/2}\right)\frac{\tilde{u}_e \tau_w}{2Pr_w} \tag{33}$$

Similarly, we replace the heat transfer term $q_{1/2}$ of the energy equation in the CFD viscous subroutine by q_w, as shown in Fig. 16. The relation between $q_{1/2}$ and q_w can be obtained from the heat transfer formula in the near-wall region proposed by Gao et al. [13,22]

$$q_{1/2} = q_w - \frac{r\tilde{u}_{1/2}\tau_w}{Pr_t} \tag{34}$$

where $\tilde{u}_{1/2}$ needs to be calculated according to the velocity law of the wall Eq. (31).

It should be emphasized that when iteratively solving u_τ from Eq. (31), the μ_w and λ_w of the gas mixture need to be calculated by Wilke's mixing law [27]. Moreover, the Pr_w in Eq. (33) should be calculated from the wall variables of the gas mixture

$$Pr_w = \frac{\mu_w c_{pw}}{\lambda_w} \tag{35}$$

instead of being regarded as a constant of 0.72.

Finally, the turbulent transport variable and eddy viscosity at the first grid point off the wall should also be corrected when using the wall function method in RANS simulation due to the coarse mesh near the wall. Based on the constant shear stress assumption Eq. (20) and velocity law of the wall Eq. (31), we obtain

$$\frac{\mathrm{d}y^+}{\mathrm{d}u^+} = \frac{\mu + \mu_t}{\mu} = [1 + \omega^*\beta^* u^+ - \omega^*\Gamma^*(u^+)^2] + \frac{\mathrm{d}\hat{y}^+_{\text{white}}}{\mathrm{d}u^+} - \kappa e^{-\kappa B}\left[1 + \kappa u^+ + \frac{(\kappa u^+)^2}{2}\right] \tag{36}$$

Then, the eddy viscosity at the first grid point off the wall μ_{t1} is given as

$$\frac{\mu_{t1}}{\mu_w} = [1 + \omega^*\beta^* u^+ - \omega^*\Gamma^*(u^+)^2] + \frac{\mathrm{d}\hat{y}^+_{\text{white}}}{\mathrm{d}u^+} - \kappa e^{-\kappa B}\left[1 + \kappa u^+ + \frac{(\kappa u^+)^2}{2}\right] - \frac{\mu_1}{\mu_w} \tag{37}$$

where

$$\frac{\mathrm{d}\hat{y}^+_{\text{white}}}{\mathrm{d}u^+} = 2\hat{y}^+_{\text{white}}\frac{\kappa\sqrt{\Gamma^*}}{Q^*}\left[1 - \frac{(2\Gamma^* u^+ - \beta^*)^2}{(Q^*)^2}\right]^{-0.5} \tag{38}$$

Further, the Spalart–Allmaras turbulence model variable at the first grid point off the wall $\tilde{\nu}_{t1}$ can be calculated by solving the following nonlinear equation

$$\mu_{t1} = \bar{\rho}_1 \tilde{v}_{t1} f_{v1} \tag{39}$$

where $f_{v1} = \chi^3/(\chi^3 + c_{v1}^3)$, $\chi \equiv \tilde{v}_{t1}/v_1$, and $c_{v1} = 7.1$

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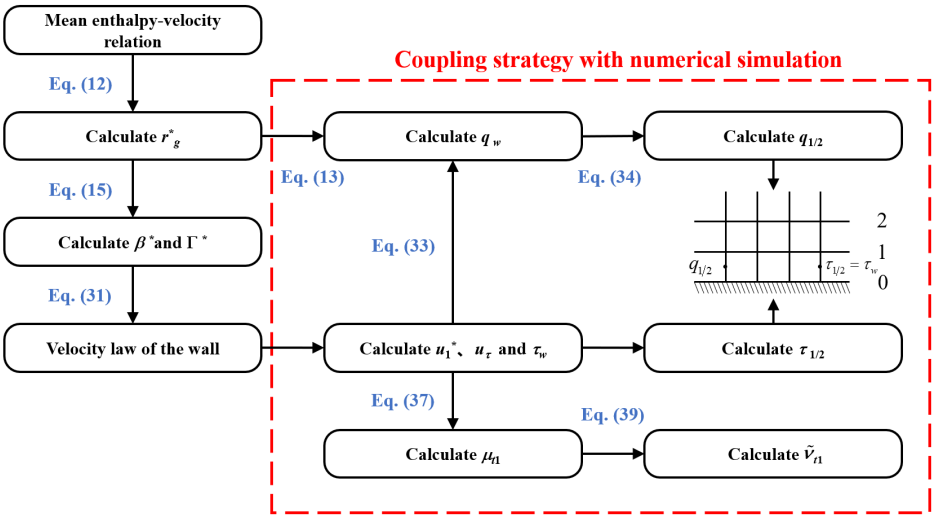


Fig. 16 Coupling strategy between the wall function method and CFD code.

B. Application of the Improved Wall Function Method

In this section, the two cases of Wedge15H25 and Wedge20H25 are employed to carry out the application studies on the improved wall function method for HRTBLs. The computational meshes and boundary conditions for numerical simulations are similar to those in Sec. III.A, except for the distribution of the near-wall grids and Δy_1. Besides, the numerical results with the dense mesh in Sec. III.A will be used for comparison.

Firstly, two coarse meshes with $\Delta y_1 = 1.2 \times 10^{-5}$ m and $\Delta y_1 = 1.5 \times 10^{-5}$ m are generated respectively for the cases of Wedge15H25 and Wedge20H25 to examine the performance of the improved wall function method when the first grid point off the wall is relaxed to the viscous sublayer ($\Delta y_1^+ = 8$). Figures 17 and 18 show the distributions of c_f and q_w predicted by the improved wall function and the existing wall function (Gao wall function [13]), and numerical results without using the wall function method are employed for reference. It can be seen that without using the wall function method, the coarse mesh fails to accurately capture the velocity and temperature gradients in the near-wall region due to the insufficiency of grid resolution. At $x = 0.8$ m (position A), the relative errors of c_f and q_w are over 19 and 34% compared with the results of the dense mesh. Although the existing wall function slightly improves the prediction accuracy, the relative errors of c_f and q_w are still over 8.0 and 10.2%. In contrast, the results predicted by the improved wall function method are in good agreement with the results of the dense mesh, with the relative errors of c_f and q_w less than 0.4 and 0.5%, respectively.

To further verify the reliability of the improved wall function method when the near-wall grid is relaxed to the logarithmic layer, two coarse meshes with $\Delta y_1 = 2 \times 10^{-4}$ m ($\Delta y_1^+ = 104$ for Wedge15H25 and $\Delta y_1^+ = 133$ for Wedge20H25) and $\Delta y_1 = 3 \times 10^{-4}$ m ($\Delta y_1^+ = 156$ for Wedge15H25 and $\Delta y_1^+ = 200$ for Wedge20H25) are generated. At this point, it is found that, without using the wall function method, the coarse mesh can no longer provide reliable results. Moreover, at position A, the relative errors of c_f and q_w obtained by the existing wall function method are greater than 11.0 and 15.3%, whereas the improved wall function method reduces the relative errors to less than 1.2 and 5.8%, as shown in Figs. 19 and 20.

For computational efficiency, Figs. 21a and 21b compare the residuals of the coarse mesh and dense mesh for the cases of Wedge15H25 and Wedge20H25, respectively, under the same CFL condition, CFL = 0.5. If the residual drop by eight orders of magnitude is used as the convergence criterion, the dense meshes ($\Delta y_1^+ < 0.4$) need to advance 73,500 and 77,000 steps, respectively, whereas the coarse meshes need only 27,000 and 24,000 steps when using the wall function method and relaxing Δy_1 to the viscous sublayer ($\Delta y_1^+ = 8$), and the computational efficiency is enhanced by 2.7 times. Moreover, the numerical steps are reduced to 4900 and 4200 when the near-wall grid is further relaxed to the logarithmic layer (Δy_1^+ are 200 and 156, respectively), enhancing the

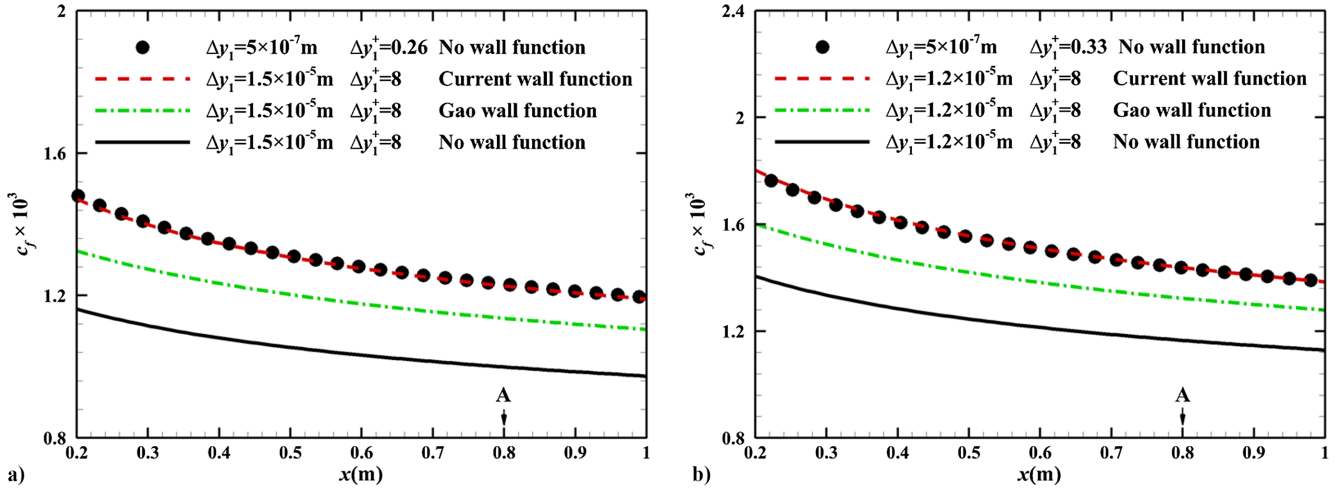


Fig. 17 Comparisons of c_f among the dense mesh, coarse mesh ($\Delta y_1^+ = 8$) with Gao wall function [13], coarse mesh with current wall function, and coarse mesh without wall function of a) Wedge15H25 and b) Wedge20H25.

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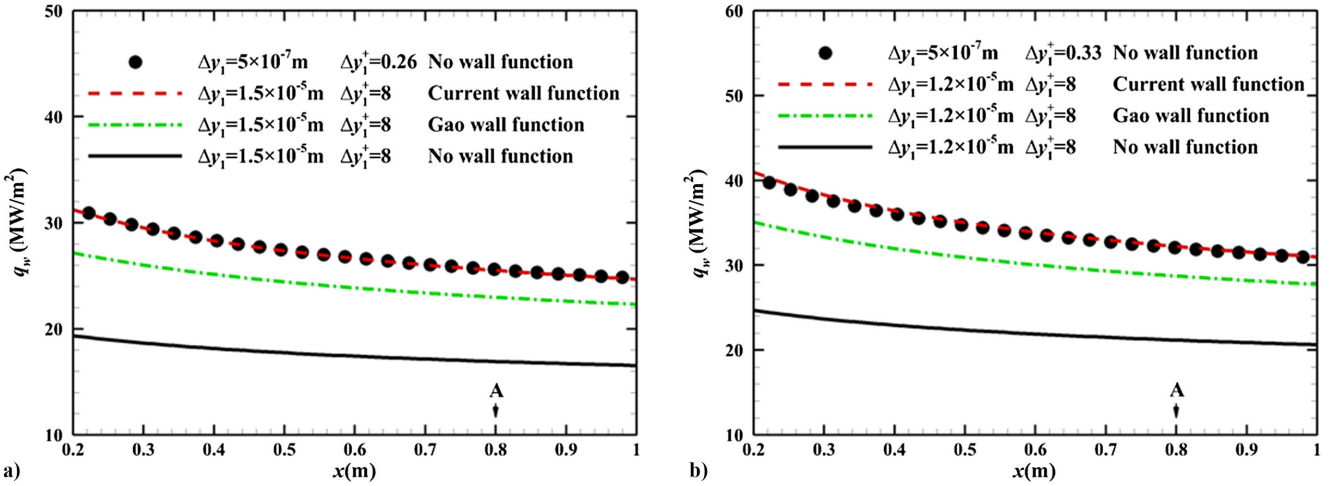


Fig. 18 Comparisons of q_w among the dense mesh, coarse mesh ($\Delta y_1^+ = 8$) with Gao wall function [13], coarse mesh with current wall function, and coarse mesh without wall function of a) Wedge15H25 and b) Wedge20H25.

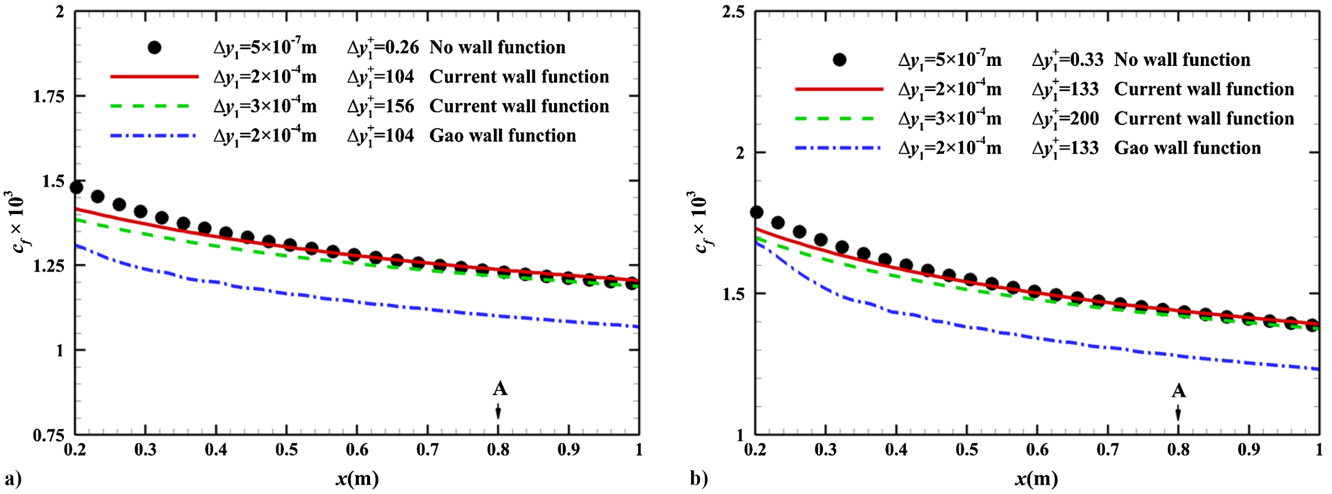


Fig. 19 Comparisons of c_f among the dense mesh, coarse mesh ($\Delta y_1 = 2 \times 10^{-4}$ m and $\Delta y_1 = 3 \times 10^{-4}$ m) with Gao wall function [13], and coarse mesh with current wall function of a) Wedge15H25 and b) Wedge20H25.

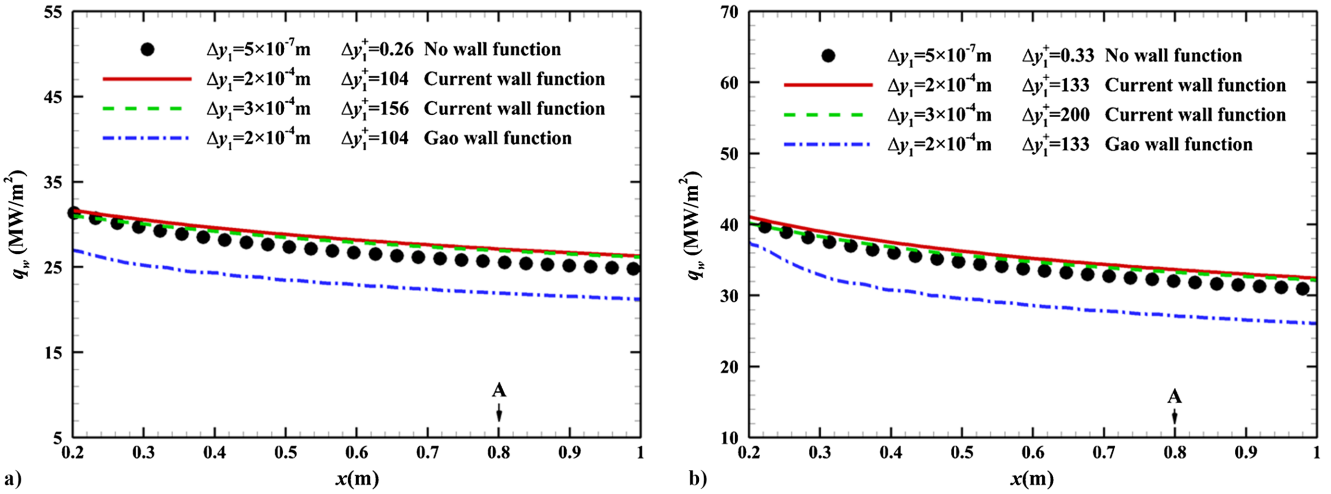


Fig. 20 Comparisons of q_w among the dense mesh, coarse mesh ($\Delta y_1 = 2 \times 10^{-4}$ m and $\Delta y_1 = 3 \times 10^{-4}$ m) with Gao wall function [13], and coarse mesh with current wall function of a) Wedge15H25 and b) Wedge20H25.

computational efficiency by 15 times. Obviously, the enlargement of Δy_1 relaxes the local timestep, significantly accelerating the simulation convergence.

The previously given numerical experiments indicate that the improved wall function method proposed in this paper effectively reduces the near-wall grid dependence of c_f and q_w in RANS

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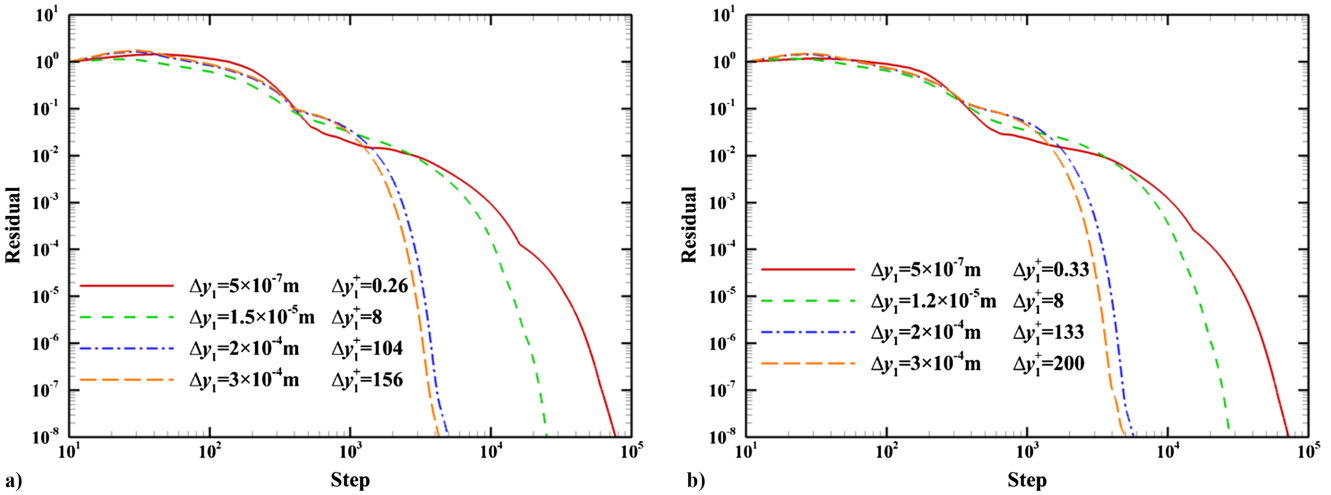


Fig. 21 Comparisons of residuals of the dense mesh and coarse mesh with current wall function of a) Wedge15H25 and b) Wedge20H25.

simulations of HRTBLs. It can predict accurate results of c_f and q_w when the first grid point off the wall is relaxed to the viscous sublayer and logarithmic layer, and the efficiency of numerical simulations is significantly enhanced, achieving fast and accurate prediction of aerodynamic/aerothermal environment of HRTBLs.

C. Method to Remove the Quantities at the Boundary Layer Edge in the Wall Function Method

From Eq. (12) and Eq. (31), it can be seen that these equations contain multiple boundary layer edge quantities (such as $\tilde{u}_e$, $\tilde{h}_e$), which are difficult to obtain in simulations of complex flow fields. This will not facilitate the applications of the wall function method. To deal with this issue, we introduce the concept of the reference point proposed by Mo et al. [43] in their study on the wall function method for hypersonic laminar boundary layers, and then the boundary layer edge quantities in the laws of the wall are replaced by those at the reference point inside the boundary layer. If the quantities at the reference point are donated as $\tilde{u}_{e-}$ and $\tilde{h}_{e-}$, the extended generalized enthalpy–velocity relation can be obtained:

$$\tilde{h} = \tilde{h}_w + (\tilde{h}_{rg} - \tilde{h}_w)\frac{\tilde{u}}{\tilde{u}_{e^-}} + (\tilde{h}_{e^-} - \tilde{h}_{rg})\left(\frac{\tilde{u}}{\tilde{u}_{e^-}}\right)^2 \tag{40}$$

where $\tilde{h}_{rg}$ and r_g* are accordingly defined as

$$\tilde{h}_{rg} = \tilde{h}_{e^-} + \frac{r_g^*\tilde{u}_{e^-}^2}{2}, \quad r_g^* = \frac{\tilde{h}_w - \tilde{h}_{e^-}}{\tilde{u}_{e^-}^2/2} + \frac{2Pr_w}{\tilde{u}_{e^-}}\frac{q_w}{\tau_w} \tag{41}$$

It can be seen from Eq. (40) that the enthalpy–velocity relation does not have to be related to the boundary layer edge quantities but only to the quantities at a reference point $e-$ within the boundary layer.

To verify the reliability of Eq. (40), Fig. 22 provides the comparisons between Eq. (40) and numerical enthalpy–velocity distributions in HRTBLs for the cases of Wedge15H25 and Wedge20H25. Here, Eq. (40) is calculated by selecting three typical positions within the HRTBLs ($\tilde{u}_{e-} = 0.1\tilde{u}_\infty$, $\tilde{u}_{e-} = 0.4\tilde{u}_\infty$, and $\tilde{u}_{e-} = 0.7\ \tilde{u}_\infty$, respectively) as the reference point. It can be seen that Eq. (40) collapses well with the numerical results, indicating that Eq. (40) successfully avoids introducing the boundary layer edge quantities while still ensuring the accuracy of the results.

Moreover, the new unified velocity law of the wall Eq. (31) can be further extended to be

$$y^+ = u^+\left[1 + \frac{\omega^*\beta^*}{2}u^+ - \frac{\omega^*\Gamma^*}{3}(u^+)^2\right] + \hat{y}^+_{\text{white}} - e^{-\kappa B}\left[1 + \kappa u^+ + \frac{(\kappa u^+)^2}{2} + \frac{(\kappa u^+)^3}{6}\right] \tag{42}$$

where the quantities at the reference point are used to replace those at the boundary layer edge in the definitions of β^* and Γ^*.

After that, in practical applications, the second point off the wall is taken as the reference point in the wall function method because the selection of the reference point is arbitrary in the wall-normal direction within HRTBLs. At this point, the coupling strategy between the wall function method and CFD code is similar to that in Sec. IV.A,

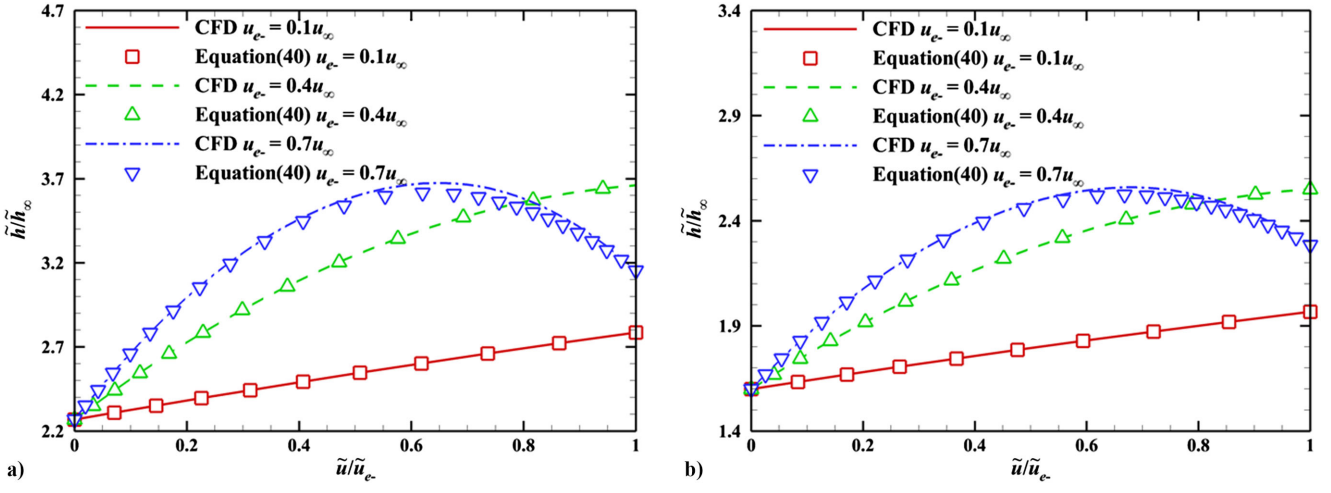


Fig. 22 Comparisons between Eq. (40) and numerical results for the cases of a) Wedge15H25 and b) Wedge20H25.

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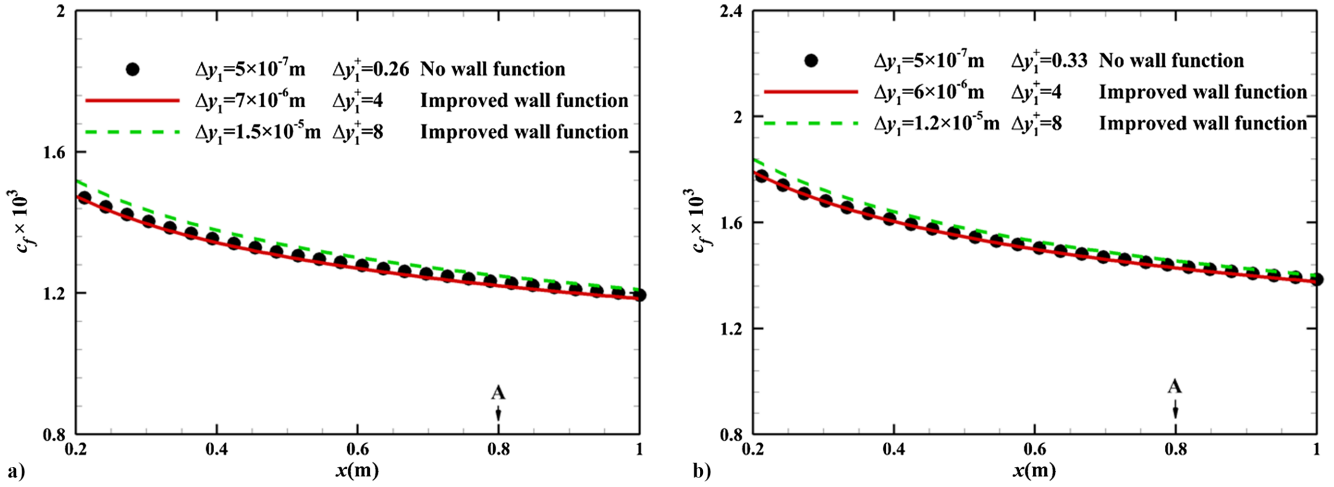


Fig. 23 Comparisons of c_f between the dense mesh and the coarse mesh with the improved wall function (based on the reference point) of a) Wedge15H25 and b) Wedge20H25.

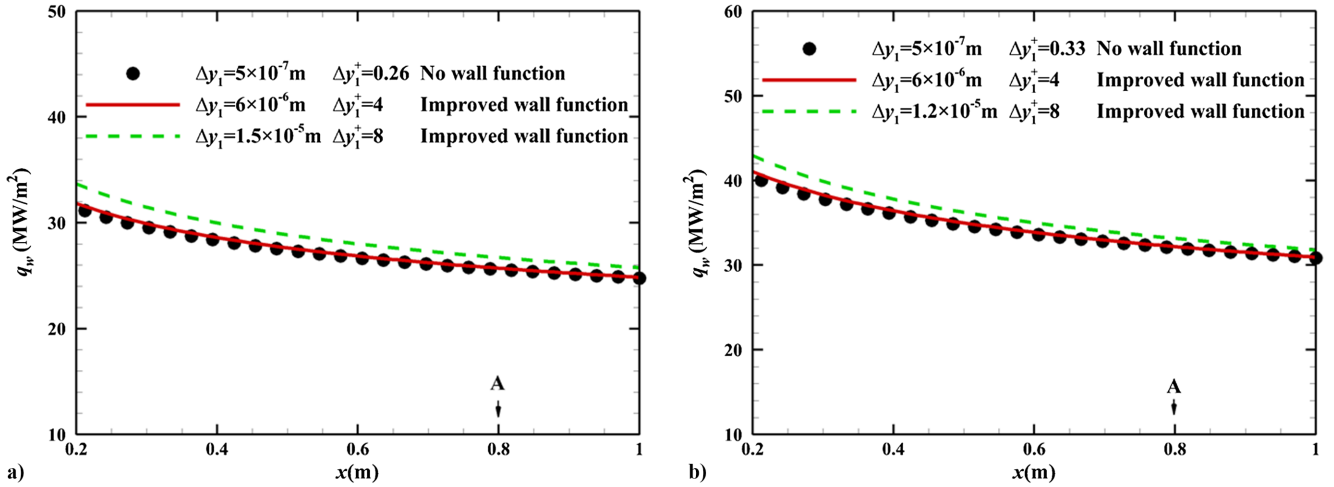


Fig. 24 Comparisons of q_w between the dense mesh and the coarse mesh with the improved wall function (based on the reference point) of a) Wedge15H25 and b) Wedge20H25.

except that the quantities at the boundary layer edge are replaced by those at the reference point when solving r_g^*, β^*, and q_w.

Next, the improved wall function method based on the reference point is applied to simulations of typical HRTBLs. To verify the reliability of the improved wall function method when the near-wall grid is relaxed to the viscous sublayer ($\Delta y_1^+ = 4$ and $\Delta y_1^+ = 8$), two coarse meshes with $\Delta y_1 = 7 \times 10^{-6}$ m and $\Delta y_1 = 1.5 \times 10^{-5}$ m are generated for Wedge15H25, and two coarse meshes with $\Delta y_1 = 6 \times 10^{-6}$ m and $\Delta y_1 = 1.2 \times 10^{-5}$ m are generated for Wedge20H25. Figures 23 and 24 compare the distributions of c_f and q_w between the dense mesh without wall function and the coarse mesh with the improved wall function (based on the reference point). It can be seen that when the first grid point off the wall is relaxed to $\Delta y_1^+ = 4$, the numerical results obtained by the current wall function method collapse well with the results of the dense mesh, with the relative errors of c_f and q_w less than 0.8 and 0.4% at $x = 0.8$ m (position A), respectively. Moreover, the relative errors of c_f and q_w obtained by the current wall function method are less than 1.4 and 4.3% when the near-wall grid is further relaxed to $\Delta y_1^+ = 8$. These results indicate that by introducing the reference point, the improved wall function method successfully removes the boundary layer edge quantities while ensuring the accuracy of numerical simulations.

V. Conclusions

The chemical reaction and severe cold wall effects in HRTBLs will cause the failure of the existing compressible wall function method. To deal with this issue, new analytic laws of the wall involving the chemical reaction and severe cold wall effects are theoretically derived under the noncatalytic wall condition. Based on these theoretical derivations, an improved wall function method is designed for efficient and accurate predictions of skin friction c_f and wall heat flux q_w in RANS simulation of HRTBLs with the one-temperature model. The main conclusions of this paper are as follows:

1) The Walz temperature–velocity relation used in the existing wall function method fails in HRTBLs. The temperature–velocity relation based on the generalized recovery factor developed by Zhang et al. is introduced to involve the cold wall effect, and this relation is further extended to a generalized enthalpy–velocity relation to consider the chemical reaction effect. Numerical experiments show that the new enthalpy–velocity relation accurately describes the correlation between static enthalpy and velocity in the inner layer of HRTBLs under the chemical reaction and severe cold wall conditions.

2) The viscous sublayer velocity law of the wall used in the existing wall function method shows significant deviation from numerical results under the chemical reaction and severe cold wall conditions due to ignoring the variation of the molecular viscosity. By establishing a new viscosity–enthalpy power law and performing Taylor expansion, a new analytic velocity law of the wall for the viscous sublayer is obtained, and the numerical experiments show it significantly reduces the prediction errors of the velocity from 19.7% to less than 2.2%. For the logarithmic layer, based on the derivation of the White compressible velocity law of the wall, the generalized enthalpy–velocity relation is introduced to replace the Walz relation,

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and the resulting velocity law of the wall reduces the prediction errors of the velocity from 4.2% to less than 1.0%. Finally, a unified form of the velocity law of the wall for the inner layers of HRTBLs is obtained by combining the two velocity laws of the wall for the viscous sublayer and logarithmic layer.

3) Based on the newly derived enthalpy–velocity relation and unified form of velocity law of the wall, an improved wall function method for RANS simulation of HRTBLs is designed, and its coupling strategy with CFD code is further established. The application studies of the wall function method for typical HRTBLs show that compared with the existing wall function method, the improved wall function method can significantly reduce the relative errors of c_f and q_w from 8.0% and 10.2% to 0.4% and 0.5%, respectively, when the first grid point off the wall is relaxed to the viscous sublayer ($\Delta y_1^+ = 8$), leading to 2.7 times enhancement of the simulation efficiency compared with the dense mesh ($\Delta y_1^+ < 0.4$); moreover, the relative errors of c_f and q_w will be reduced from 11.0% and 15.3% to 1.2% and 5.8%, respectively, when the near-wall grid is further relaxed to the logarithmic layer (Δy_1^+ are 200 and 156, respectively), leading to 15 times enhancement of the simulation efficiency compared with the dense mesh. Furthermore, by introducing the reference point, the improved wall function method successfully removes the boundary layer edge quantities while ensuring the accuracy of numerical simulations.

Appendix A

We present here the verification of grid independence for the cases in Sec. III.A. The coarse, medium, and dense meshes with $\Delta y_1 = 3 \times 10^{-6}$ m, $\Delta y_1 = 5 \times 10^{-7}$ m, and $\Delta y_1 = 3 \times 10^{-7}$ m are generated for the cases of Wedge15H25 and Wedge20H25. The computational meshes and boundary conditions for numerical simulations are similar to those in Sec. III.A, except for the distribution of the near-wall grids and Δy_1. Figures A1 and A2 summarize the numerical results obtained by different meshes. It can be seen that for both of the cases, the results obtained by the medium mesh are almost identical to those of the dense mesh, indicating that the medium mesh is enough to achieve grid independence. Therefore, it is reasonable to use the medium mesh to perform the numerical simulations in Sec III.A.

Acknowledgments

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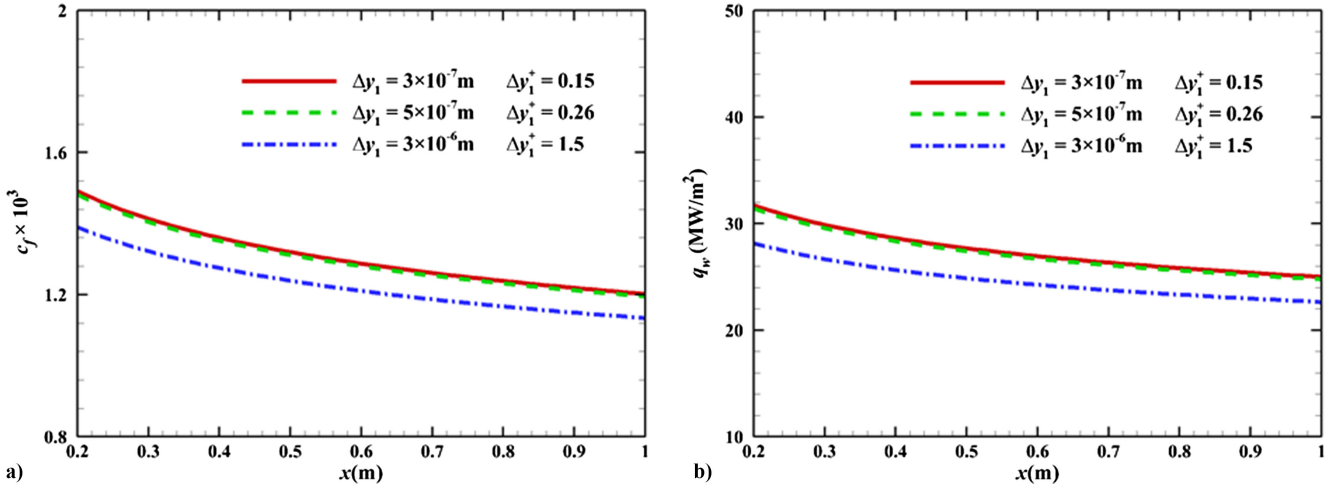


Fig. A1 Comparisons of a) c_f and b) q_w among the coarse, medium, and dense meshes for the case of Wedge15H25.

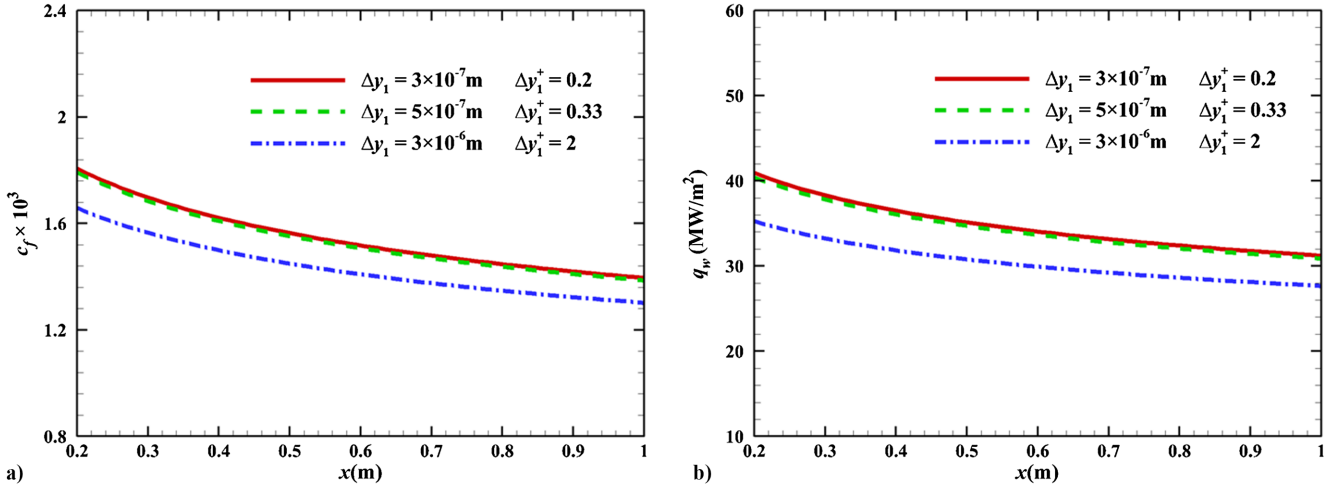


Fig. A2 Comparisons of a) c_f and b) q_w among the coarse, medium, and dense meshes for the case of Wedge20H25.

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